Chapter 7: Integration
Part I

Calculus II

April 20, 2025

Table of Contents

1. Volume and Cavalieri's principle
2. The double integral over a rectangle
3. The double integral over regions
4. Change of variables in a double integral

Section 1

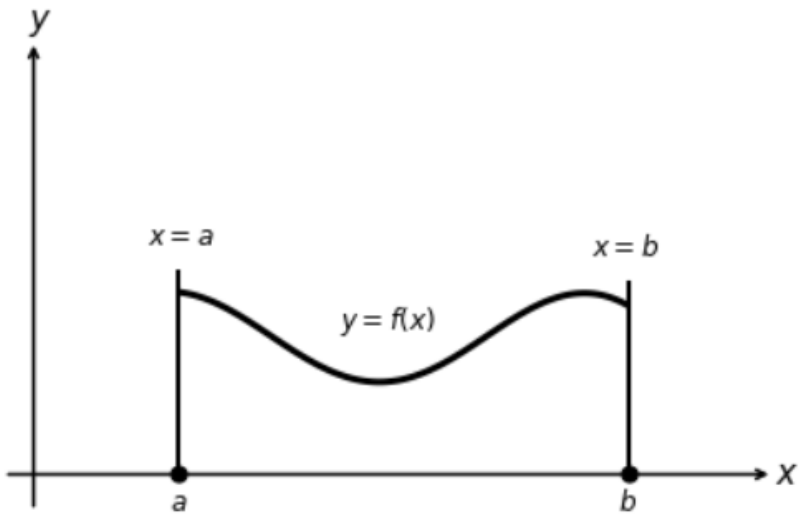
Volume and Cavalieri's principle

Geometric Interpretation of the Integral

From **one-variable calculus**: if f is nonnegative, the integral

$$\int_a^b f(x)\,\mathrm{d}x$$

equals the area of the region under the graph of f on $[a, b]$.



- If the function f is both negative and positive, then the integral is interpreted as a **signed area**, with the areas below the x axis being counted negative.

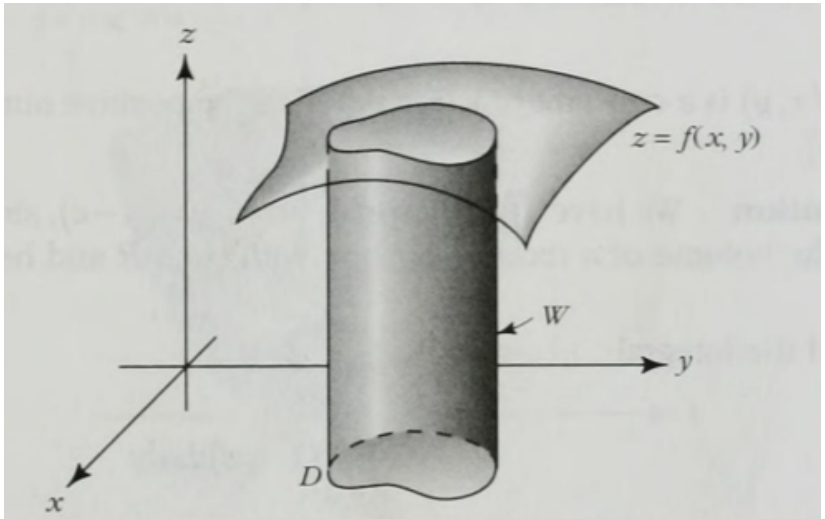
Double integral and volume

The operation of double integration assigns to a function $f(x, y)$ defined on a region D in the plane a number

$$\iint_D f(x, y)\, \mathrm{d}x\mathrm{d}y, \quad \text{also denoted} \quad \iint_D f(x, y)\, \mathrm{d}A,$$

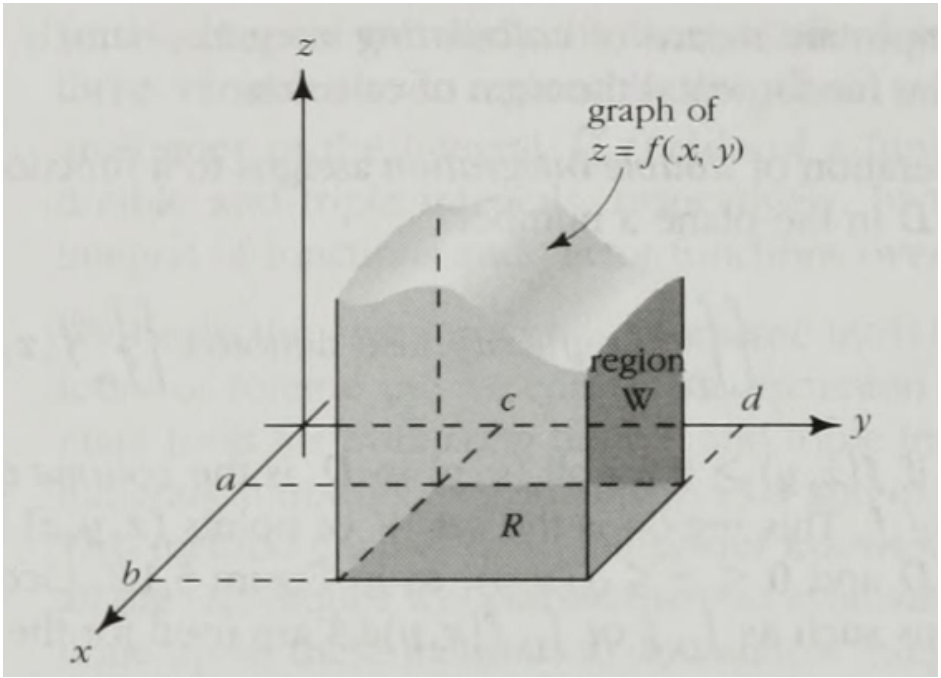
which, if $f(x, y) \geq 0$ for all (x, y) in D, is the **volume of the region under the graph of** f.

- This region is the set W of points (x, y, z) in $\mathbb{R}^3$ for which (x, y) lies in D and $0 \leq z \leq f(x, y)$.



Continue

When $D = R$ is a rectangle whose sides are parallel to the x and y axes, the solid region W appears as below.



Note: The rectangle R is determined by the two closed intervals $[a, b]$ and $[c, d]$, representing the sides of R along the x and y axes, respectively.

We say that R is the **Cartesian product** of $[a, b]$ and $[c, d]$ and write

$$R = [a, b] \times [c, d]$$

Example 7.1.1

Find

$$\iint_R f(x,y)\ \mathrm{d}A$$

If $R = [a,b] \times [c,d]$ and $f(x,y)$ is a constant function equal to the positive number k.

Solution:

We have

$$\iint_R f(x,y)\ \mathrm{d}A = k(b-a)(d-c),$$

since the integral is equal to the volume of a rectangular box with base R and height k.

Example 7.1.2

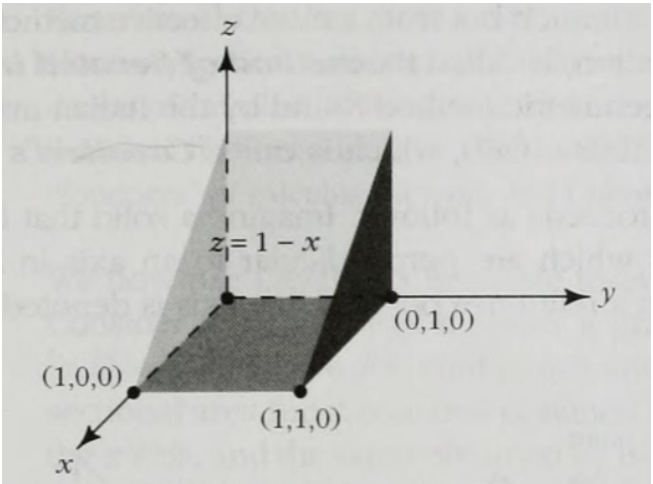
Find the integral

$$\iint_R (1-x)\,\mathrm{d}x\mathrm{d}y$$

where the region R is $[0,1]\times[0,1]$.

Solution:

The integral equals the volume of the wedge-shaped region shown below.



This region is half of a unit cube, so

$$\iint_R (1-x)\,\mathrm{d}x\mathrm{d}y = \frac{1}{2}$$

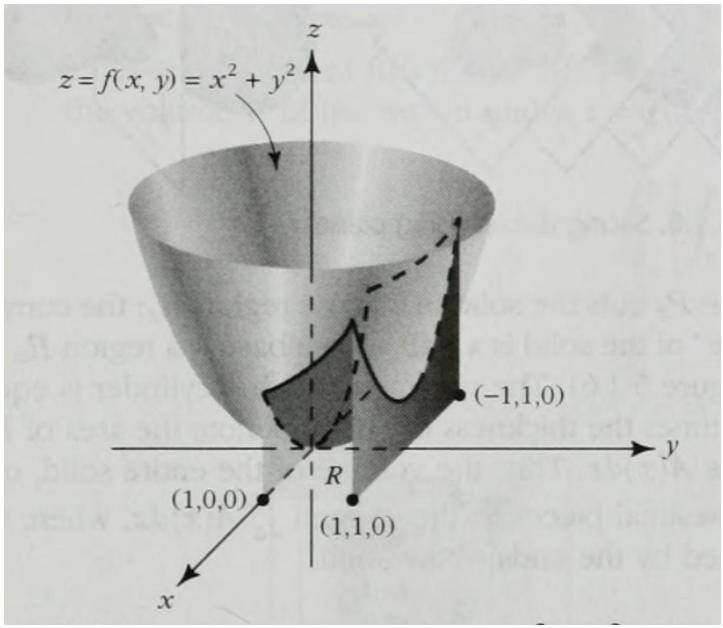
Example 7.1.3

Sketch the solid whose volume is

$$\iint_R (x^2 + y^2)\ \mathrm{d}x\mathrm{d}y$$

where the region R is $[-1, 1] \times [0, 1]$.

The solid is shown below.



There does not seem to be a straightforward way to compute its volume as in the previous two examples.

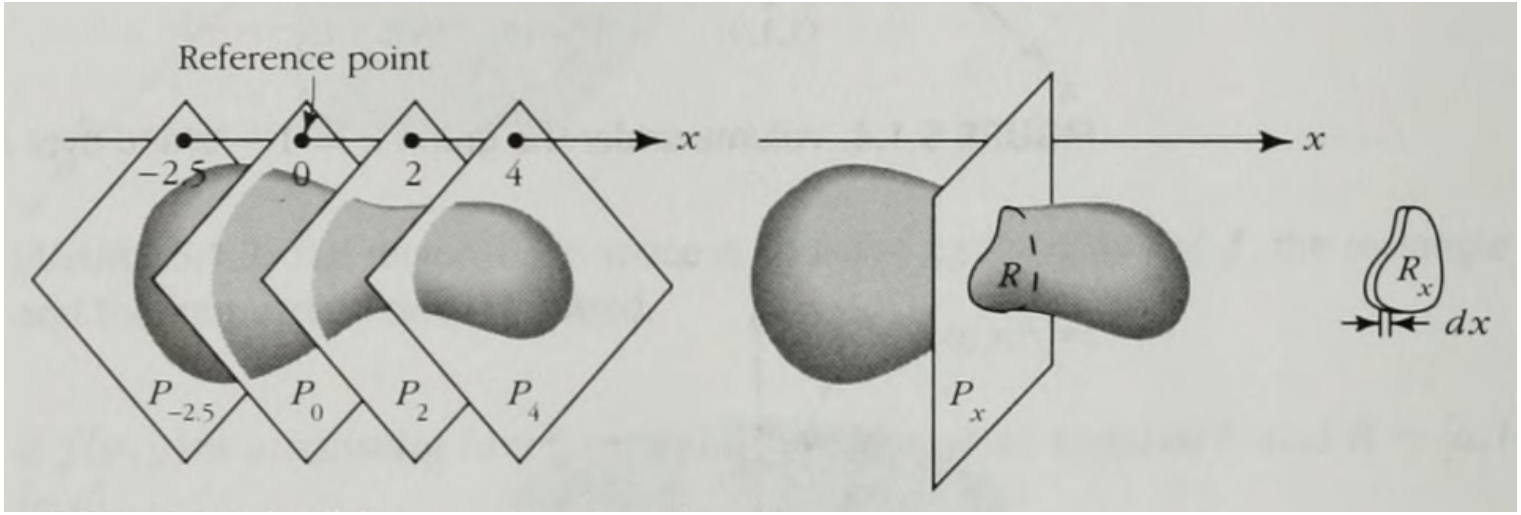
Strategy to compute volume

As with areas in single-variable calculus, the volume of W is not computed directly from its definition but from a more effective method which, in the case of multiple integration, is called the **method of iterated integrals**.

- This method is based on a geometric intuition found by the Italian mathematician Bonaventura Cavalieri (1598-1647), which is now called **Cavalieri's principle**.

Cavalieri's Principle

Imagine a solid that is sliced by a family of parallel planes which are perpendicular to an axis in space.
The plane at a distance x from a reference point on the axis is denoted P_x.



- The plane P_x cuts the solid in a plane region R_x
- The corresponding "infinitesimal piece" of the solid is a slab whose
  - Base is a region R_x
  - Thickness is $\mathrm{d}x$

The volume of such a cylinder is equal to the area of the base R_x times the thickness $\mathrm{d}x$.

Continue

If we denote the area of R_x by $A(x)$, then this volume is $A(x)\ \mathrm{d}x$.

Thus the volume of the entire solid, obtained by summing the infinitesimal pieces, is the integral $\int_a^b A(x)\ \mathrm{d}x$, where the limits a and b are determined by the ends of the solid.

Definition 7.1.4 (The Slice Method - Cavalieri's Principle)

Let S be a solid and P_x be a family of parallel planes such that:

1. S lies between P_a and P_b
2. The area of the slice of S cut by P_z is $A(x)$

Then the volume of S is equal to

$$\int_a^b A(x)\ \mathrm{d}x$$

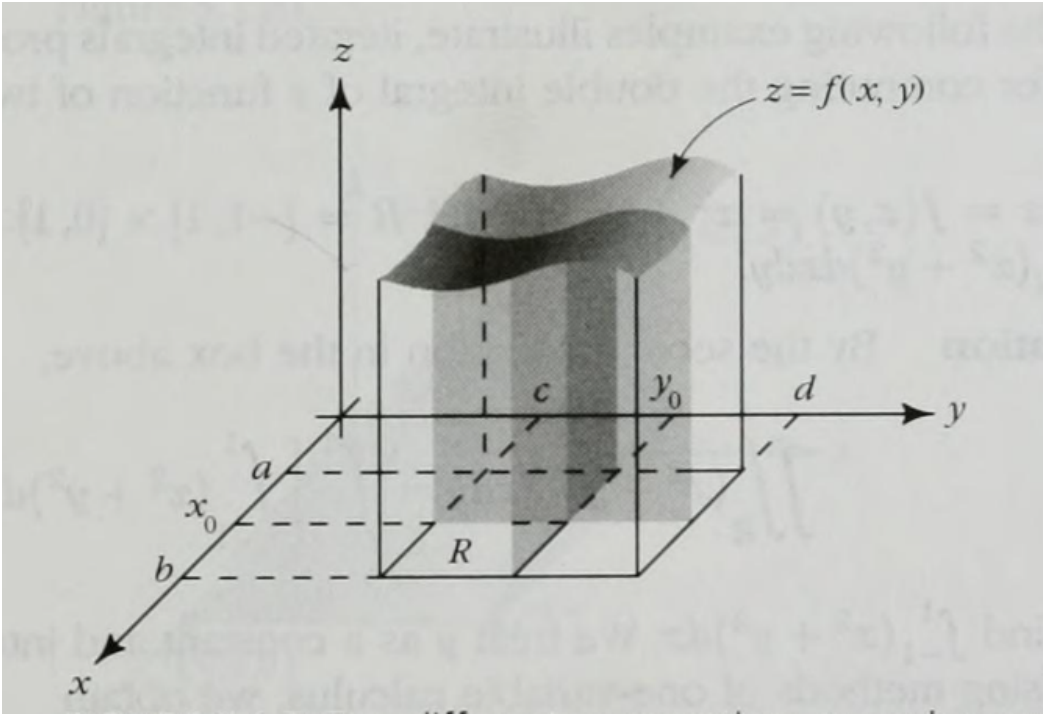
Cavalieri's principle: evaluate double integrals over rectangles

Consider the solid region under a graph $z = f(x, y)$ defined on the region $[a, b] \times [c, d]$, where $f > 0$ is continuous.

When cutting the solid with planes, there are two ways:

- **Cutting planes perpendicular to the x-axis**, fixing $x = x_0$.
- **Cutting planes perpendicular to the y-axis**, fixing $y = y_0$.

Continue



The cross section determined by a cutting plane $x = x_0$ is the plane region under the graph of $z = f(x_0, y)$ from $y = c$ to $y = d$.

When we fix $x = x_0$, the cross-sectional area $A(x_0)$ is the integral $\int_c^d f(x_0, y)\ \mathrm{d}y$. Thus the cross-sectional area function is

$$A(x) = \int_c^d f(x, y)\ \mathrm{d}y$$

Continue

By Cavalieri's principle, the volume V of the region under $z = f(x, y)$ equals

$$V = \int_a^b A(x)\,\mathrm{d}x = \int_a^b \left[\int_c^d f(x, y)\,\mathrm{d}y\right]\mathrm{d}x.$$

The integral $\int_a^b \left[\int_c^d f(x, y)\,\mathrm{d}y\right]\mathrm{d}x$ is known as an **iterated integral**, because it is obtained by carrying out the one-variable integration process twice:

1. Ignore the outer integral and evaluate $\int_c^d f(x, y)\,\mathrm{d}y$ to get a function of x alone.
2. Integrate this function of x from a to b to obtain a number.

Continue

Since $\iint_R f(x,y)\ \mathrm{d}A$ is equal to the volume V, we get the following identity; it turns out to be valid even when f is allowed to take on negative values.

Theorem 7.1.5 (Double and Iterated Integrals)

If f is a continuous function over a rectangle $R = [a,b] \times [c,d]$,

$$\iint_R f(x,y)\ \mathrm{d}A = \int_a^b \left[\int_c^d f(x,y)\ \mathrm{d}y \right] \mathrm{d}x$$

If we reverse the roles of x and y in the above discussion and use cutting planes perpendicular to the y axis, we obtain

$$\iint_R f(x,y)\ \mathrm{d}A = \int_c^d \left[\int_a^b f(x,y)\ \mathrm{d}x \right] \mathrm{d}y \tag{7.1}$$

Example 7.1.6

Let $R = [-1, 1] \times [0, 1]$, and

$$z = f(x, y) = x^2 + y^2$$

Evaluate the integral

$$\iint_R \left(x^2 + y^2\right) \, \mathrm{d}x\mathrm{d}y$$

Solution

1. By equation (7.1),

$$\iint_R \left(x^2+y^2\right)\mathrm{d}x\mathrm{d}y = \int_0^1 \left[\int_{-1}^1 \left(x^2+y^2\right)\ \mathrm{d}x\right]\mathrm{d}y$$

2. To find $\int_{-1}^1 \left(x^2+y^2\right)\mathrm{d}x$, we treat y as a constant and integrate with respect to x:

$$\int_{-1}^1 \left(x^2+y^2\right)\mathrm{d}x = \left[\frac{x^3}{3}+y^2x\right]_{x=-1}^1 = \frac{2}{3}+2y^2$$

3. Integrate $\frac{2}{3}+2y^2$ with respect to y from 0 to 1, to obtain

$$\int_0^1 \left(\frac{2}{3}+2y^2\right)\mathrm{d}y = \left[\frac{2}{3}y+\frac{2}{3}y^3\right]_{y=0}^1 = \frac{4}{3}$$

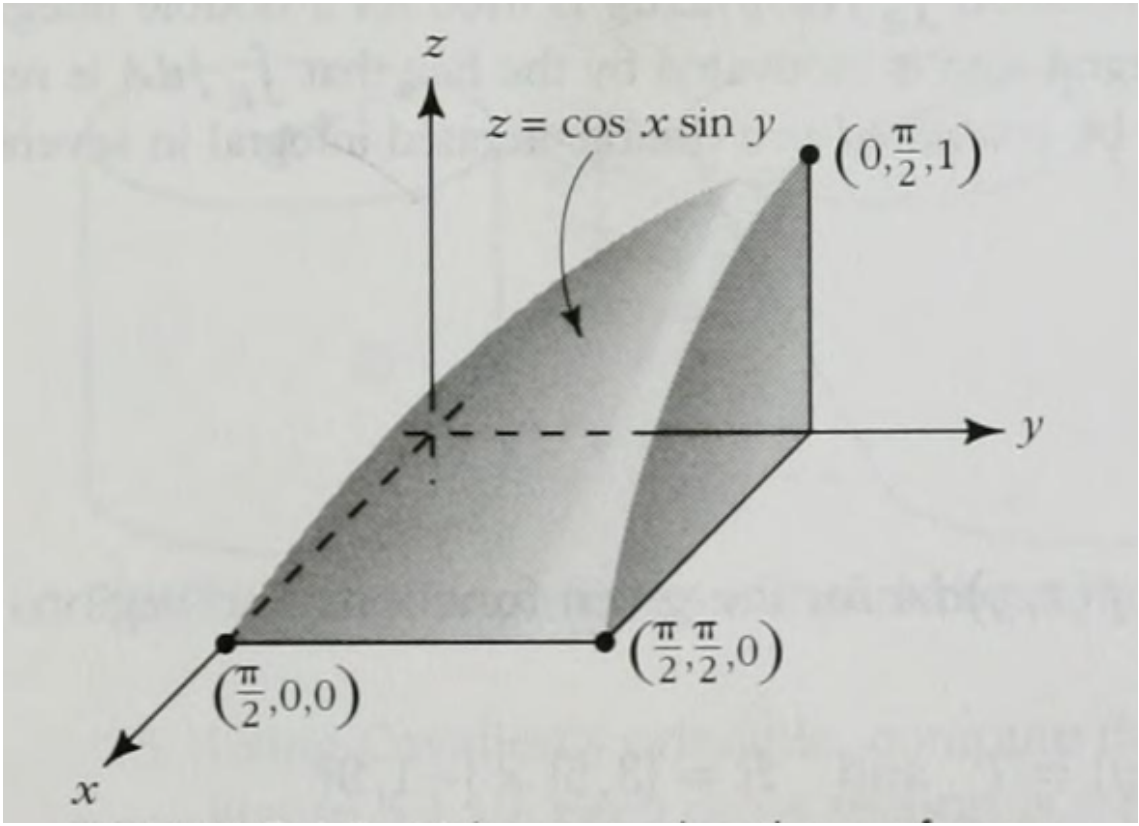
Hence the volume of the solid is $4/3$.

Example 7.1.7

Compute

$$\iint_S \cos x \sin y \, \mathrm{d}x\mathrm{d}y$$

where S is the square $[0, \pi/2] \times [0, \pi/2]$.



Solution

By equation (7.1), we have:

$$\begin{aligned}\iint_S \cos x \sin y \, \mathrm{d}x\mathrm{d}y &= \int_0^{\pi/2} \left[\int_0^{\pi/2} \cos x \sin y \, \mathrm{d}x \right] \mathrm{d}y \\ &= \int_0^{\pi/2} \sin y \left[\int_0^{\pi/2} \cos x \, \mathrm{d}x \right] \mathrm{d}y \\ &= \int_0^{\pi/2} \sin y \, \mathrm{d}y = 1.\end{aligned}$$

Remark

It is common to delete the brackets in iterated integrals and write

$$\int_a^b \int_c^d f(x,y)\ \mathrm{d}y\mathrm{d}x \quad \text{instead of} \quad \int_a^b \left[\int_c^d f(x,y)\ \mathrm{d}y\right] \mathrm{d}x$$

and

$$\int_c^d \int_a^b f(x,y)\ \mathrm{d}x\mathrm{d}y \quad \text{instead of} \quad \int_c^d \left[\int_a^b f(x,y)\ \mathrm{d}x\right] \mathrm{d}y$$

Section 2

The double integral over a rectangle

Definition vs. Calculation

In the previous section, our treatment of double integrals was based on the geometric notion of volume. Since multiple integration

1. has many applications besides this geometric one
2. needs to handle cases where function values can be positive or negative
3. needs to clearly define the "volume" of a complicated region

Therefore, a more general definition is required.

Single-Variable Case

- We do not use the definition to calculate integrals.
  - Rather, we use anti-differentiation and the fundamental theorem of calculus.

Double Integrals

- Similarly, the techniques for calculating are those illustrated in the last section:
  - Reduction to iterated integrals
  - Combined with one-variable techniques

Continue

Question: Why is the "technical" definition given below necessary at all, when all calculations are done by a simpler method?

An answer to this question is that the definitions help us to **understand** the fundamentals of the integration process.

- For example, many integrals are difficult or even impossible to evaluate by antidifferentiation methods.

  Such integrals can be estimated, however, by numerical methods, many of which are based on these definitions.

Recall: In single-variable calculus, to define $\int_a^b f(x)\,\mathrm{d}x$, we start with subdividing an interval into smaller subintervals.

Definition 7.2.1

The cartesian product $R = [a, b] \times [c, d]$ of two closed intervals is called a **rectangle**.

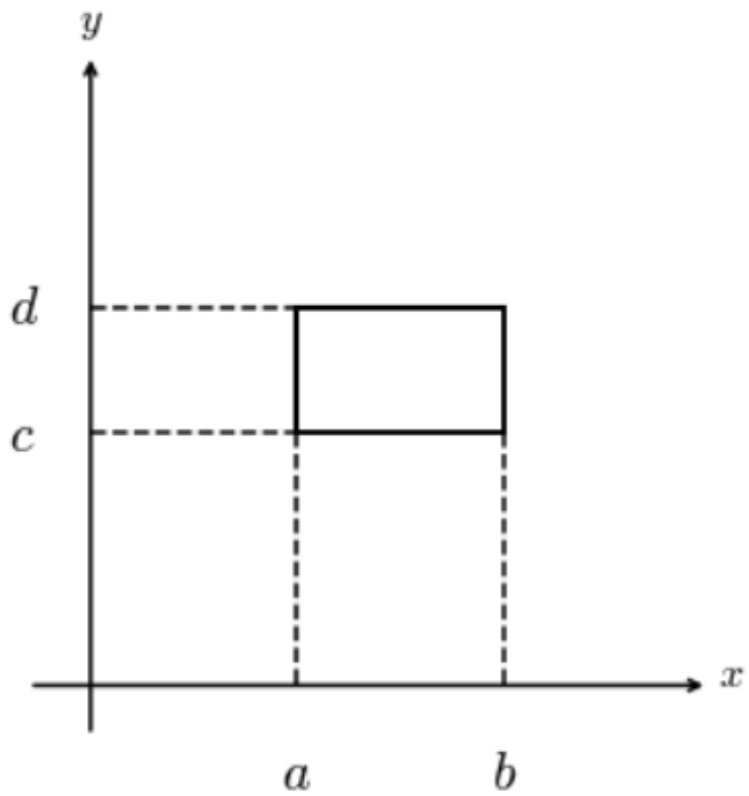
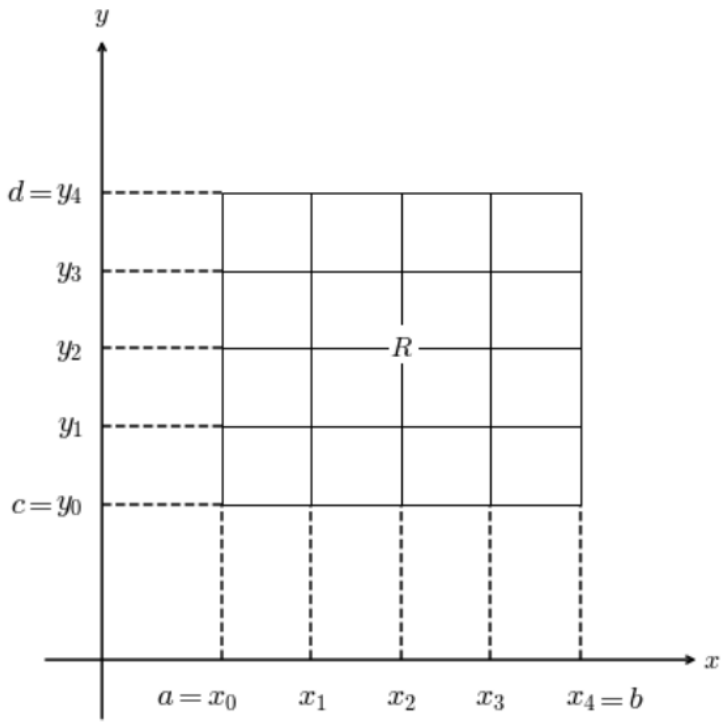


Figure 7.1: $R = [a, b] \times [c, d]$ consists of (x, y)'s such that $a \leq x \leq b$ and $c \leq y \leq d$.



Definition 7.2.2

A **regular partition** of R of order n consists of:

- Two collections of $n+1$ equally spaced points:

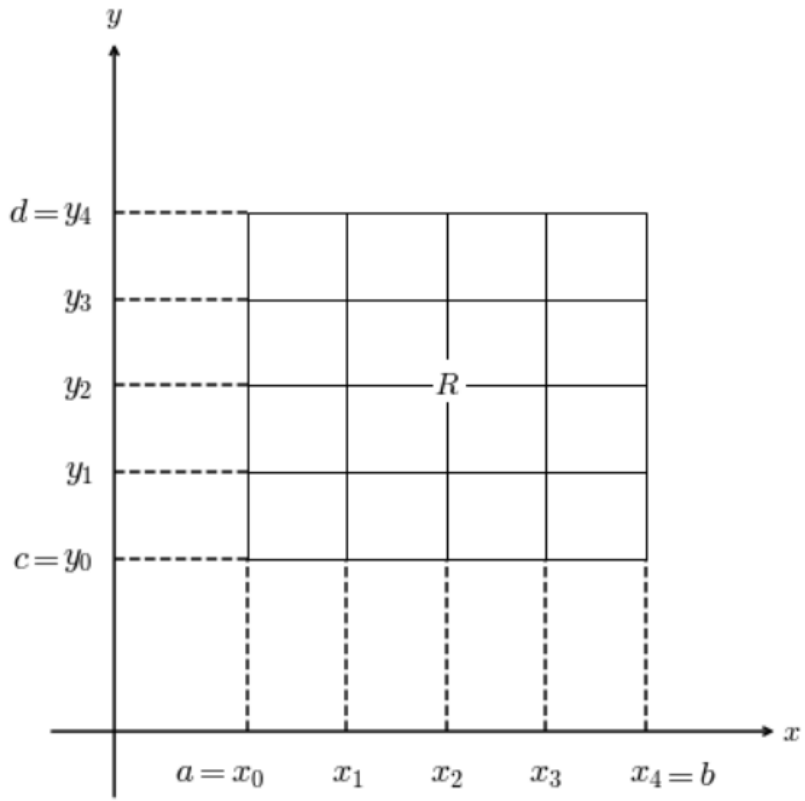
$$\{x_j\}_{j=0}^{n} : a = x_0 < x_1 < \cdots < x_n = b$$
$$\{y_k\}_{k=0}^{n} : c = y_0 < y_1 < \cdots < y_n = d$$

- With uniform spacing:

$$\Delta x = x_{j+1} - x_j = \frac{b-a}{n},$$
$$\Delta y = y_{k+1} - y_k = \frac{d-c}{n}$$

Remark: One could modify the condition that the points are equally spaced, but it would complicate the exposition a little.

Label the smaller rectangles in a partition



For given j and k, let

$$R_{jk} := [x_j, x_{j+1}] \times [y_k, y_{k+1}]$$

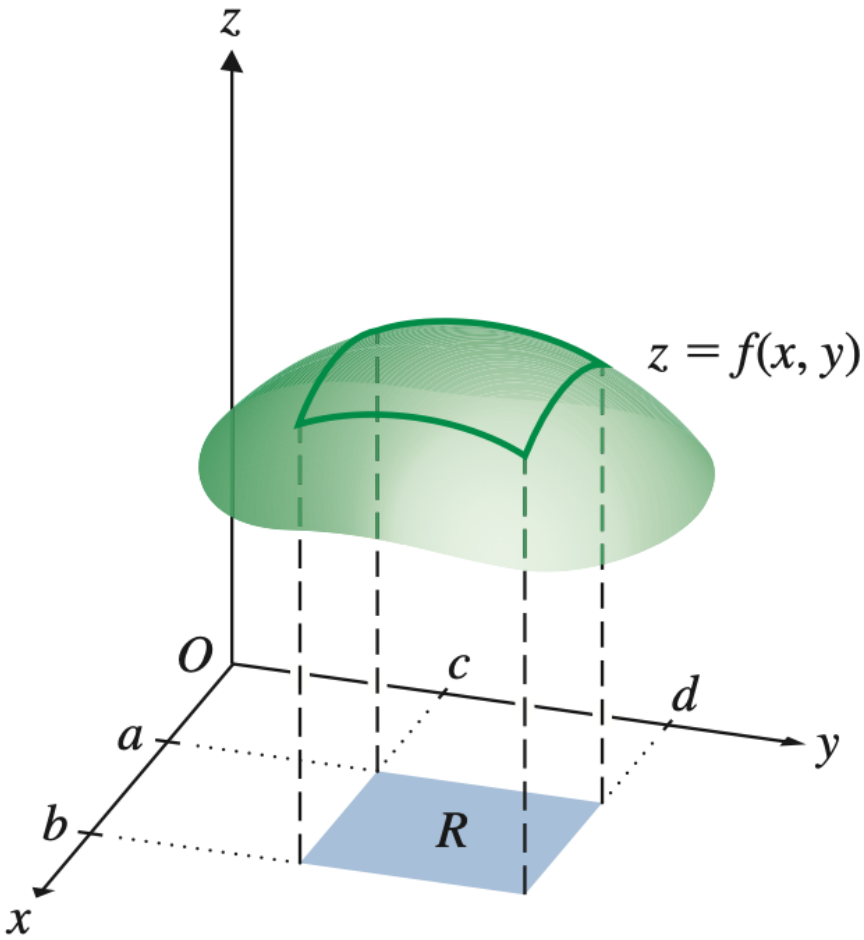


Figure 7.2: Volume under the surface $z = f(x, y)$

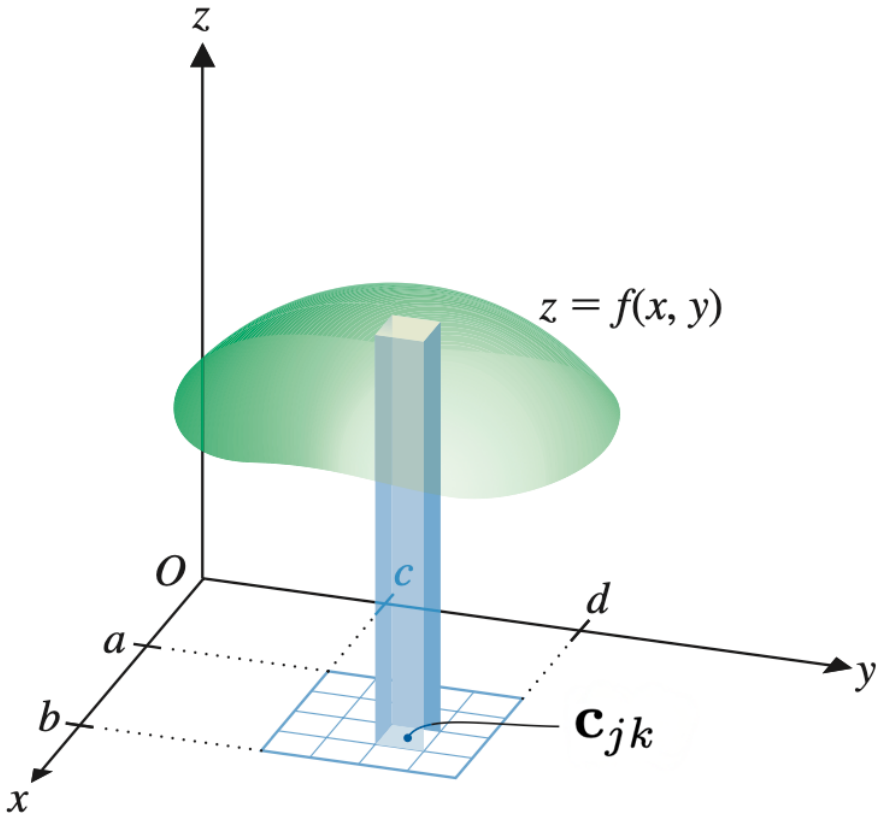


Figure 7.3: Approximating the volume above R_{jk} by a rectangular box

Volume of the rectangular box:

$$f\left(\mathbf{c}_{jk}\right)\Delta x\Delta y$$

Definition 7.2.3 (Riemann sum)

Suppose f is a bounded real-valued function defined on R. Let $\mathbf{c}_{jk}$ be any point in R_{jk}.

We call

$$S_n = \sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\,\Delta x \Delta y = \sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\,\Delta A$$

a Riemann sum, where

$$\Delta x = x_{j+1} - x_j = \frac{b-a}{n}, \quad \Delta y = y_{k+1} - y_k = \frac{d-c}{n}, \quad \Delta A = \Delta x \Delta y.$$

Note that the above Riemann sum is taken over all j's and k's from 0 to $n-1$, and so there are n^2 terms.

Definition of the Double Integral

Definition 7.2.4

If the sequence of Riemann sums $\{S_n\}$ converges to a limit S as $n \to \infty$ and the limit S is the same for any choice of points $\mathbf{c}_{jk}$ in the rectangles R_{jk}, then we say that f is **integrable** over R and we write

$$\iint_R f(x,y)\ \mathrm{d}A \quad \text{or} \quad \iint_R f(x,y)\ \mathrm{d}x\mathrm{d}y$$

for the limit S.

Thus we can describe integrability in the following way:

$$\lim_{n\to\infty} \sum_{j,k=0}^{n-1} f\left(\mathbf{c}_{jk}\right) \Delta x \Delta y = \iint_R f(x,y)\ \mathrm{d}A$$

for any choice of $\mathbf{c}_{jk} \in R_{jk}$.

Geometry of the Double Integral

If $f(x,y) \geq 0$, the existence of $\lim_{n\to\infty} S_n$ has a straightforward geometric meaning. Consider

- Graph of $z = f(x,y)$ as top surface
- Rectangle R as the base of the solid

Inscribed Solid

- Choose $\mathbf{c}_{jk}$ where f has its **minimum** on R_{jk}
- Volume element: $f(\mathbf{c}_{jk})\Delta x\Delta y$
- Total volume: $\sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\Delta x\Delta y$

Circumscribed Solid

- Choose $\mathbf{c}_{jk}$ where f has its **maximum** on R_{jk}
- Volume element: $f(\mathbf{c}_{jk})\Delta x\Delta y$
- Total volume: $\sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\Delta x\Delta y$

Therefore, if $\lim_{n\to\infty} S_n$ exists and is independent of $\mathbf{c}_{jk} \in R_{jk}$, then the volumes of the inscribed and circumscribed solids approach the same limit as $n \to \infty$.

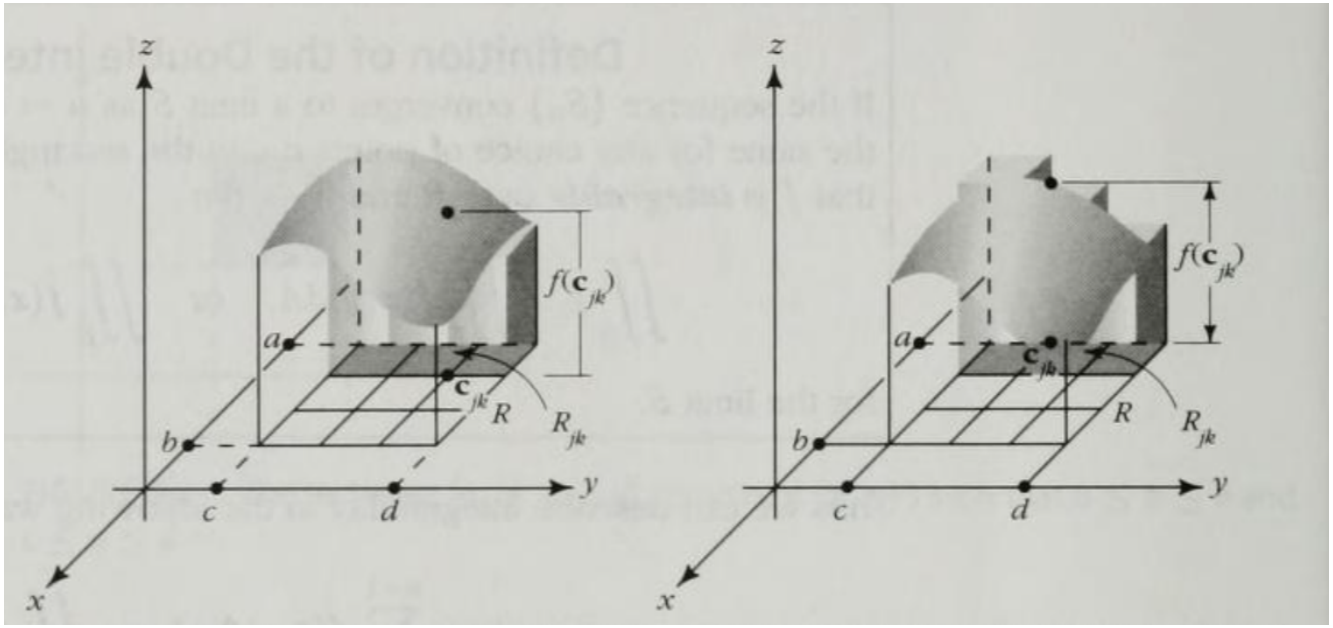


Figure 7.4: The volume of inscribed and circumscribed boxes approximates the volume under $z = f(x, y)$

Remark

If $f(x, y) \geq 0$, then

$$\iint_R f(x, y)\, \mathrm{d}x\mathrm{d}y$$

is the volume of the region under the graph of f and above the rectangle R in the xy plane.

Integrability

It is usually difficult to prove from the definition that a function is integrable.

- Existence theorem (from advanced courses):
  - Guarantees integrability for continuous functions
  - Applies to certain discontinuous functions
  - Particularly relevant for functions with discontinuities along curves in the xy-plane

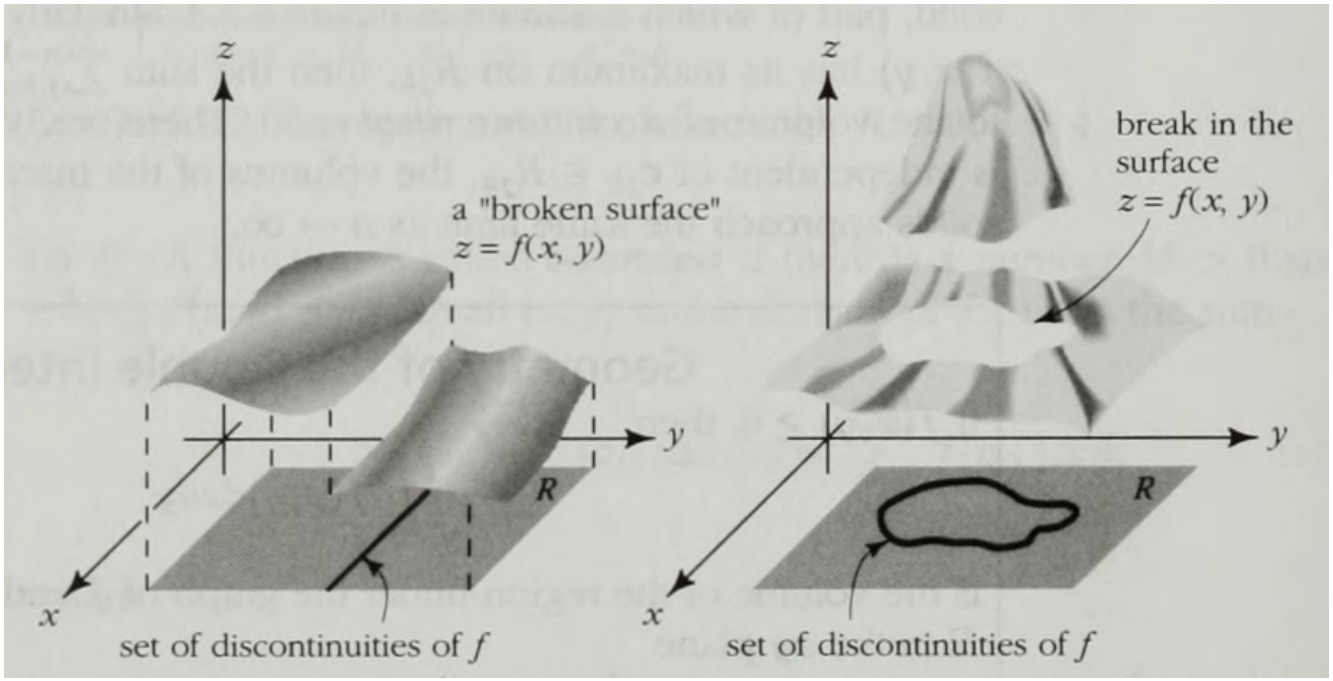
Integrability

The figure below shows two functions defined on a rectangle R whose discontinuities lie along curves. In other words, f is continuous at each point that is in R but not on the curve.

Useful curves are graphs of functions

$$y = \phi(x),\ a \le x \le b, \quad \text{or} \quad x = \psi(y),\ c \le y \le d,$$

or finite unions of such graphs.



Existence of Integrals

Theorem 7.2.5

If a function f is bounded on R and is continuous except possibly for discontinuities along a finite number of graphs of continuous functions, then f is integrable on R.

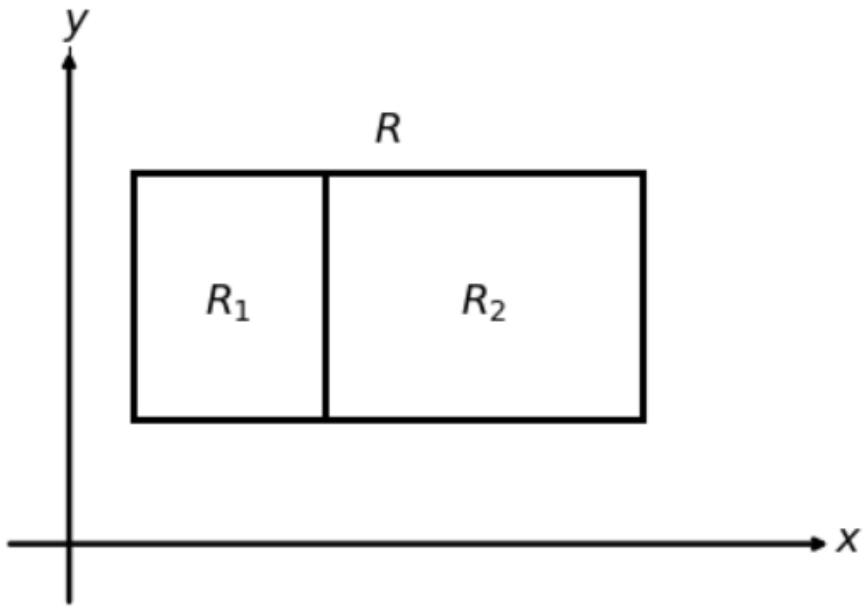
- A continuous function on a closed rectangle is always bounded,
  - but, for example, $f(x, y) = y + 1/x$ on $(0, 1] \times [0, 1]$ is not bounded, because $1/x$ becomes arbitrarily large for x near 0.

Properties of the Double Integral (stated without proof)

1. Every continuous function is integrable.

2. If a rectangle R is divided by a line segment into two rectangles R_1 and R_2, and if $f(x, y)$ is integrable on R_1 and R_2, then f is integrable on R and

$$\iint_R f(x,y)\,\mathrm{d}x\mathrm{d}y = \iint_{R_1} f(x,y)\,\mathrm{d}x\mathrm{d}y + \iint_{R_2} f(x,y)\,\mathrm{d}x\mathrm{d}y.$$



Continue

3. If f_1 and f_2 are integrable on R and if $f_1(x,y) \leq f_2(x,y)$ for all (x,y) in R, then
$$\iint_R f_1(x,y)\,\mathrm{d}x\mathrm{d}y \leq \iint_R f_2(x,y)\,\mathrm{d}x\mathrm{d}y$$

4. If $f(x,y) = k$ for all (x,y) in R,
$$\iint_R f(x,y)\,\mathrm{d}x\mathrm{d}y = k\ (\text{area of } R).$$

5.
$$\iint_R [f_1(x,y) + f_2(x,y)]\,\mathrm{d}x\mathrm{d}y = \iint_R f_1(x,y)\,\mathrm{d}x\mathrm{d}y + \iint_R f_2(x,y)\,\mathrm{d}x\mathrm{d}y$$

Continue

6.
$$\iint_R cf(x,y)\ \mathrm{d}x\mathrm{d}y = c\iint_R f(x,y)\ \mathrm{d}x\mathrm{d}y.$$

7.
$$\left|\iint f(x,y)\ \mathrm{d}x\mathrm{d}y\right| \leq \iint_R |f(x,y)|\ \mathrm{d}x\mathrm{d}y.$$

Method of Iterated Integration

Many double integrals can be computed using the iterated integration method.

Recall:
In **Section 1** we used Cavalieri's principle (slicing by planes perpendicular to the x or y axis) to show that the volume of the region under a graph can be evaluated as an iterated integral.

- A similar result can be proved, without recourse to geometry, for the integrals of functions which may take both positive or negative values.

Continue

The notation $\iint_R f(x,y)\,\mathrm{d}x\mathrm{d}y$ is used to denote the double integral, whereas

$$\int_c^d \left[\int_a^b f(x,y)\,\mathrm{d}x\right]\mathrm{d}y \quad \text{or} \quad \int_a^b \left[\int_c^d f(x,y)\,\mathrm{d}y\right]\mathrm{d}x$$

are used for iterated integrals. We emphasize that:

- Each iterated integral is the result of a calculation that involves **two one-variable integrations**
- Whereas the double integral is a quantity that "involves both variables at once."

Thus, the **reduction to iterated integrals** is a significant result which equates essentially different concepts.

- Without a result like this together with the fundamental theorem of calculus, we would not be able to evaluate double integrals.

Reduction to Iterated Integrals

Theorem 7.2.6 (Fubini)

If f is continuous on the rectangle $R = [a, b] \times [c, d]$, then both the iterated integrals

$$\int_c^d \left[\int_a^b f(x, y) \, \mathrm{d}x \right] \mathrm{d}y \quad \text{and} \quad \int_a^b \left[\int_c^d f(x, y) \, \mathrm{d}y \right] \mathrm{d}x$$

exist and equal the double integral

$$\iint_R f(x, y) \, \mathrm{d}x\mathrm{d}y.$$

As a consequence, we see that

$$\int_c^d \left[\int_a^b f(x, y) \, \mathrm{d}x \right] \mathrm{d}y = \int_a^b \left[\int_c^d f(x, y) \, \mathrm{d}y \right] \mathrm{d}x$$

Example 7.2.7

Compute

$$\iint_R (x^2 + y)\ \mathrm{d}A$$

where R is the square $[0, 1] \times [0, 1]$.

Solution:

By reduction to iterated integrals,

$$\iint_R (x^2 + y)\ \mathrm{d}A = \int_0^1 \int_0^1 (x^2 + y)\ \mathrm{d}x\mathrm{d}y = \int_0^1 \left[\int_0^1 (x^2 + y)\ \mathrm{d}x \right] \mathrm{d}y$$

The x integration is performed first:

$$\int_0^1 (x^2 + y)\ \mathrm{d}x = \left[\frac{x^3}{3} + yx \right]_{x=0}^1 = \frac{1}{3} + y$$

Thus

$$\iint_R (x^2 + y)\ \mathrm{d}A = \int_0^1 \left[\frac{1}{3} + y \right] \mathrm{d}y = \left[\frac{1}{3}y + \frac{y^2}{2} \right]_0^1 = \frac{5}{6}$$

Example 7.2.8

A consequence of the reduction to iterated integrals is that interchanging the order of integration in the iterated integrals does not change the answer.

Verify this for Example 7.2.7.

Solution:

We carry out the integration in the other order:

$$\int_0^1 \int_0^1 \left(x^2 + y\right)\ \mathrm{d}y\mathrm{d}x = \int_0^1 \left[x^2 y + \frac{y^2}{2}\right]_{y=0}^1\ \mathrm{d}x = \int_0^1 \left[x^2 + \frac{1}{2}\right]\ \mathrm{d}x$$

$$= \left[\frac{x^3}{3} + \frac{x}{2}\right]_0^1 = \frac{5}{6}$$

Example 7.2.9

Verify that

$$\int_0^2 \int_1^3 x^2 y \, dy dx = \int_1^3 \int_0^2 x^2 y \, dx dy$$

Solution:

$$\int_0^2 \int_1^3 x^2 y \, dy dx = \int_0^2 \left(\int_1^3 x^2 y \, dy \right) dx = \int_0^2 \left(\frac{x^2 y^2}{2} \Big|_{y=1}^3 \right) dx$$

$$= \int_0^2 x^2 \left(\frac{9}{2} - \frac{1}{2} \right) dx = 4 \int_0^2 x^2 \, dx = 4 \frac{x^3}{3} \Big|_0^2 = \frac{32}{3}$$

In the other order,

$$\int_1^3 \int_0^2 x^2 y \, dx dy = \int_1^3 \left(\int_0^2 x^2 y \, dx \right) dy = \int_1^3 \left(\frac{x^3 y}{3} \Big|_{x=0}^2 \right) dy$$

$$= \int_1^3 \frac{8}{3} y dy = \frac{4}{3} y^2 \Big|_1^3 = \frac{4}{3} \left(3^2 - 1^2 \right) = \frac{32}{3}$$

Example 7.2.10

Compute

$$\iint_R \sin(x+y)\ \mathrm{d}x\mathrm{d}y$$

where $R = [0,\pi] \times [0,2\pi]$.

Solution:

$$\begin{aligned}
\iint_R \sin(x+y)\ \mathrm{d}x\mathrm{d}y &= \int_0^{2\pi} \left[\int_0^{\pi} \sin(x+y)\ \mathrm{d}x \right] \mathrm{d}y \\
&= \int_0^{2\pi} [-\cos(x+y)|_{x=0}^{\pi}]\,\mathrm{d}y \\
&= \int_0^{2\pi} [\cos y - \cos(y+\pi)]\mathrm{d}y \\
&= [\sin y - \sin(y+\pi)]|_{y=0}^{2\pi} = 0
\end{aligned}$$

Example 7.2.11

Find the volume under the graph of $f(x, y) = x^2 + y^2$ between the planes $x = 0$, $x = 3$, $y = -1$, and $y = 1$.

Solution:

The volume is

$$\int_{-1}^{1}\int_{0}^{3} \left(x^2 + y^2\right)\ \mathrm{d}x\mathrm{d}y = \int_{-1}^{1} \left(\frac{x^3}{3} + y^2 x\Big|_{x=0}^{3}\right) \mathrm{d}y = \int_{-1}^{1} \left(9 + 3y^2\right) \mathrm{d}y$$

$$= \left(9y + y^3\right)\Big|_{-1}^{1} = 20$$

Example 7.2.12

Compute

$$\int_D xy^2 \, \mathrm{d}A$$

where D is the rectangle given by

$$2 \leq x \leq 3, \quad 0 \leq y \leq 1$$

Solution:

$$\int_D xy^2 \, \mathrm{d}A = \int_2^3 \left(\int_0^1 xy^2 \, \mathrm{d}y \right) \mathrm{d}x = \int_2^3 \left[\frac{1}{3} xy^3 \right]_{y=0}^1 \mathrm{d}x = \int_2^3 \frac{1}{3} x \, \mathrm{d}x = \frac{5}{6}$$

Integrating first with respect to x and then with respect to y we get

$$\int_D xy^2 \, \mathrm{d}A = \int_0^1 \left(\int_2^3 xy^2 \, \mathrm{d}x \right) \mathrm{d}y = \int_0^1 \frac{5}{2} y^2 \, \mathrm{d}y = \frac{5}{6}$$

Section 3

The double integral over regions

Many applications involve double integrals $\iint_D f(x,y)\,\mathrm{d}x\mathrm{d}y$ over regions D that are not rectangles.

For instance, the volume of a hemisphere can be expressed as such integrals.

- We first give a definition of the double integral over a region that is not necessarily a rectangle.
- Then we show how these integrals can be evaluated by a modified form of iterated integration.

The Double Integral over a Region D

If f is a function defined on a region D in the plane, we want

$$\iint_D f(x,y)\,\mathrm{d}x\mathrm{d}y$$

to agree with our previous definition when D is a rectangle.

Note: When $f(x,y) \geq 0$ on D, this integral should be the volume of the three-dimensional region under the graph of f and above D.

Two-Step Integration Process:

1. **Function Modification:**
   - Creates a function defined on all of $\mathbb{R}^2$ that matches f on D
2. **Rectangle Integration:**
   - Select any rectangle $R \supset D$ (containing D)
   - Since $\bar{f} \equiv 0$ outside D, the choice of R doesn't affect the integral value

Continue

1. Our first step uses the "mathematician's trick" of reducing a problem to one which has already been solved.

   We replace the original function f by the function $\bar{f}$ defined on the entire plane $\mathbb{R}^2$ by

$$\bar{f}(x,y) = \begin{cases} f(x,y), & \text{if } (x,y) \in D; \\ 0, & \text{if } (x,y) \notin D \end{cases}$$

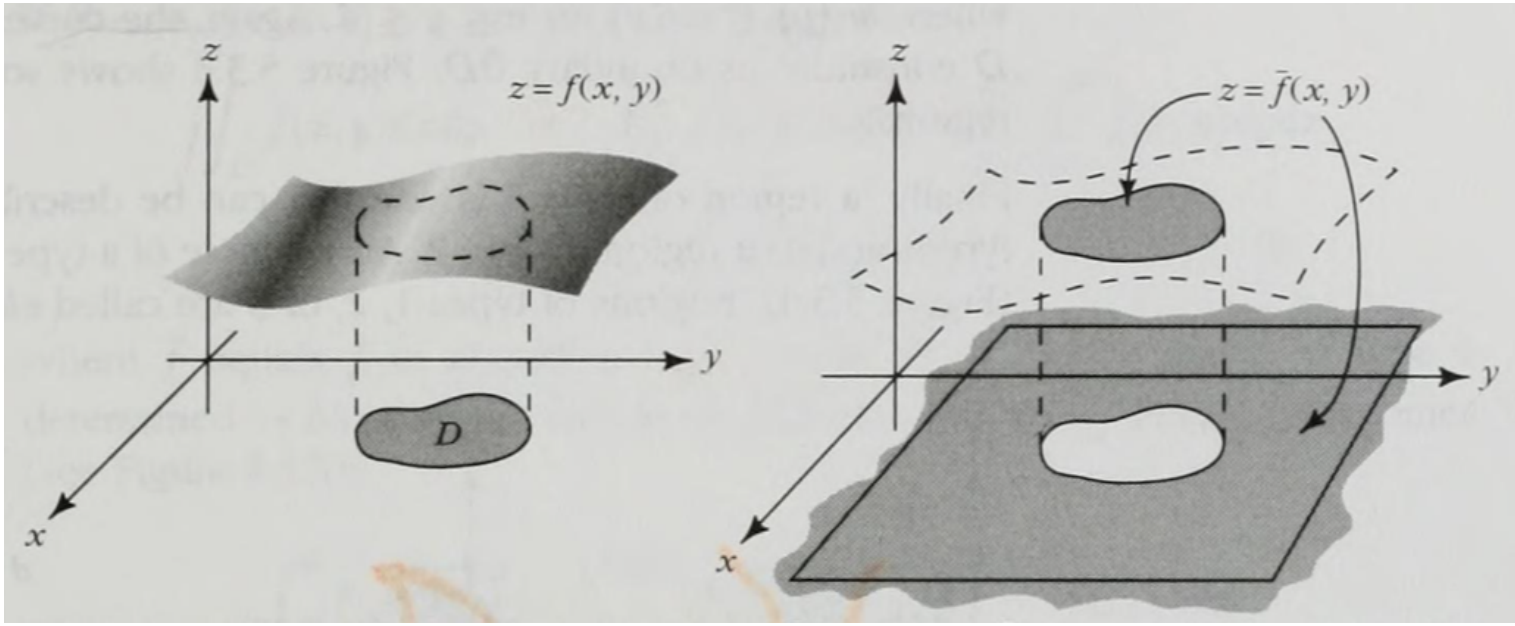


Figure 7.5: Given f and D, we construct $\bar{f}$ by setting $\bar{f}(x,y)$ equal to zero outside D.

Continue

2. The second step is to integrate the modified function $\bar{f}$ over any rectangle R that is large enough to contain D. $(R \supset D)$

   - Since $\bar{f}$ is zero outside D, it doesn't matter which such rectangle we choose, since the part of R outside D doesn't contribute to the integral.

For $f(x, y) \geq 0$ on D:

- The region W under f over D
- The region $\bar{W}$ under $\bar{f}$ over R
- Difference: $\bar{W}$ includes a flat "plate" over $R \setminus D$ at $z = 0$ (consisting of those points (x, y, z) such that (x, y) is in R but not in D, and $z = 0$)

Key observations:

- The plate contributes *zero volume* $\left(\iint_{R \setminus D} 0 \ \mathrm{d}x\mathrm{d}y = 0\right)$
- Thus,

$$\mathrm{Vol}(W) = \mathrm{Vol}(\bar{W}) = \iint_R \bar{f}(x, y) \ \mathrm{d}x\mathrm{d}y$$

Continue

Definition 7.3.1 (The Double Integral over a Region D)

Extend f to a rectangle R containing D by defining $\bar{f}(x,y)$ to be $f(x,y)$ for (x,y) in D and $\bar{f}(x,y) = 0$ outside D.

If $\bar{f}$ is integrable on R, we say that f is integrable on D, and we define

$$\iint_D f(x,y)\,\mathrm{d}x\mathrm{d}y \quad = \quad \iint_R \bar{f}(x,y)\,\mathrm{d}x\mathrm{d}y$$

Remark: This definition is **not directly useful** for evaluating integrals; however, knowing that the integral is a volume, we can evaluate a few simple ones.

Example 7.3.2

Let D be the disc $x^2 + y^2 \leq 1$, and let $f(x, y)$ be the constant function whose value is 3 for all (x, y). Find

$$\iint_D f(x, y)\, \mathrm{d}x\mathrm{d}y$$

Solution:

- The integral is the volume of the region under the graph of f on D.
- This region is a right circular cylinder with base area π and height 3, so its volume is 3π.

Hence

$$\iint_D f(x, y)\, \mathrm{d}x\mathrm{d}y = 3\pi$$

Elementary Regions for Double Integration

Definition 7.3.3 (Type 1 Regions)

For two continuous real-valued functions ϕ_1, $\phi_2 : [a, b] \to \mathbb{R}$ with $\phi_1(x) \leq \phi_2(x)$, the region:

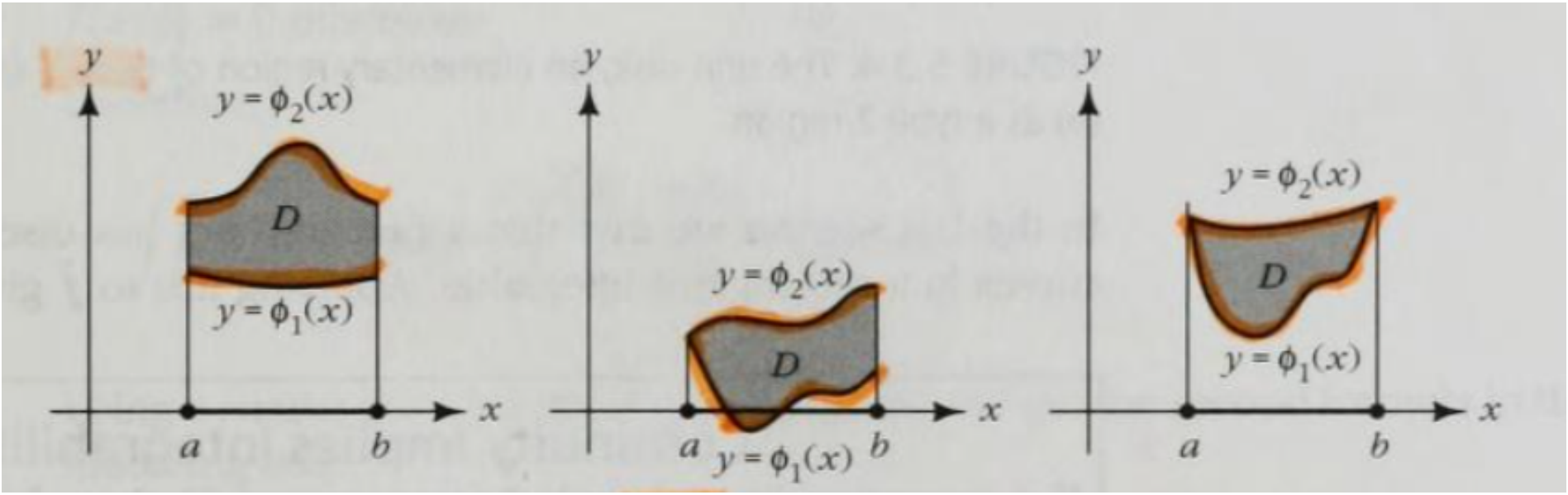
$$D = \{(x, y) \mid a \leq x \leq b,\ \phi_1(x) \leq y \leq \phi_2(x)\}$$

is called a **Type 1 elementary region**.

Type 1 elementary regions

The figure below shows various examples of regions of Type 1.

The curves and straight-line segments that bound the region constitute the boundary of D, denoted ∂D.



Definition 7.3.4 (Type 2 Regions)

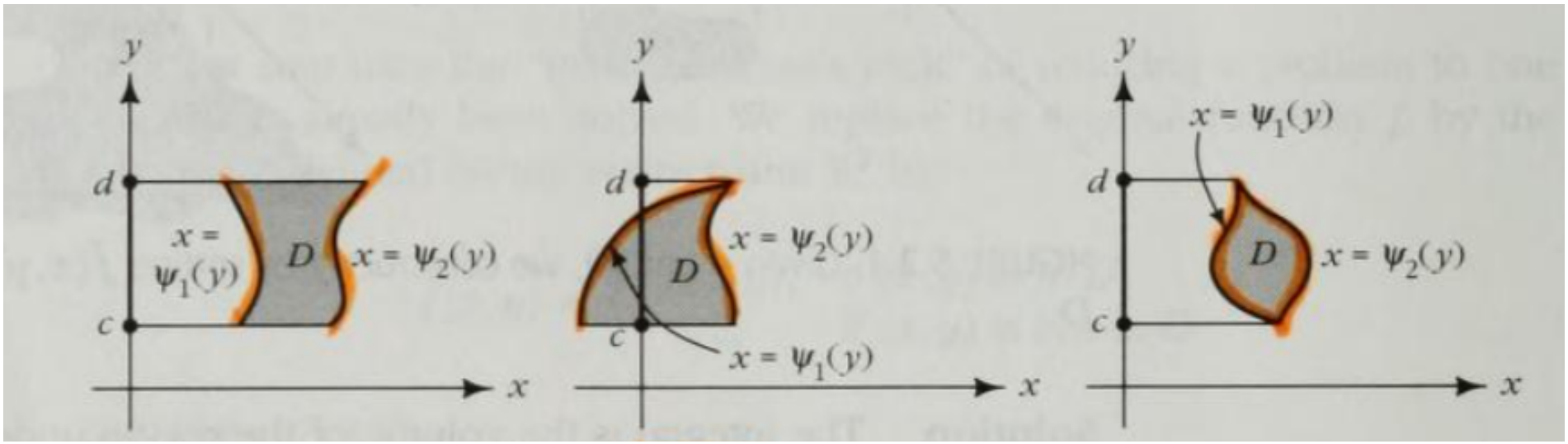
A region D is of **Type 2** if there exist continuous functions $\psi_1, \psi_2 : [c, d] \to \mathbb{R}$ such that:

$$D = \{(x, y) \mid c \leq y \leq d,\ \psi_1(y) \leq x \leq \psi_2(y)\}$$

with $\psi_1(y) \leq \psi_2(y)$ for all $y \in [c, d]$.

Again, the curves that bound the region D constitute its boundary ∂D.

The figure below shows various examples of regions of Type 2.

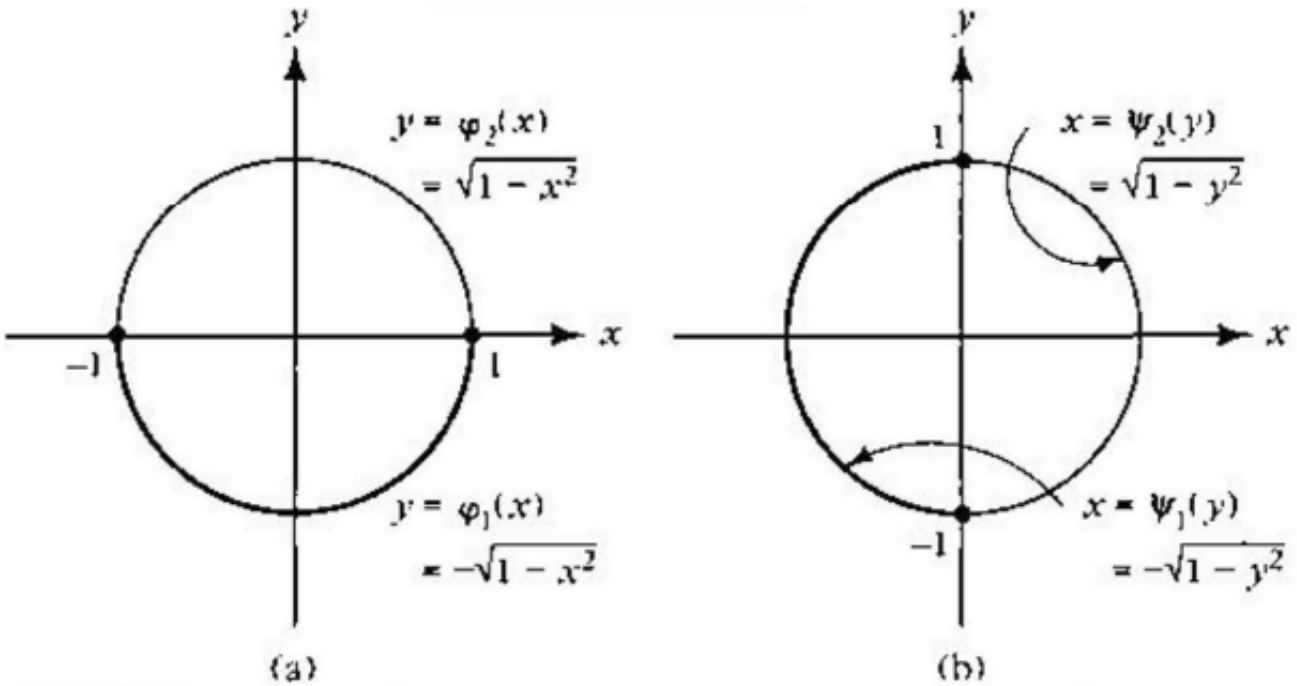


Definition 7.3.5 (Type 3 Regions)

A region is **Type 3** (elementary) if it can be described as both:

- A Type 1 region (using functions of x)
- A Type 2 region (using functions of y)

An example of a Type 3 region is the unit disk (see the figure below). Regions of Types 1, 2, or 3 are called **elementary regions**.



Integrability on Elementary Regions

In the last section, we saw that a function that has discontinuities only along curves in a rectangle is integrable. Applying this to $\bar{f}$ gives:

Theorem 7.3.6 (Continuity Implies Integrability)

If f is continuous on an elementary region D, then f is integrable on D.

This fact, combined with the reduction of iterated integrals for rectangles, enables us to evaluate integrals over elementary regions by iterated integration.

Integration Over Elementary Regions

If $R = [a, b] \times [c, d]$ is a rectangle containing D, then

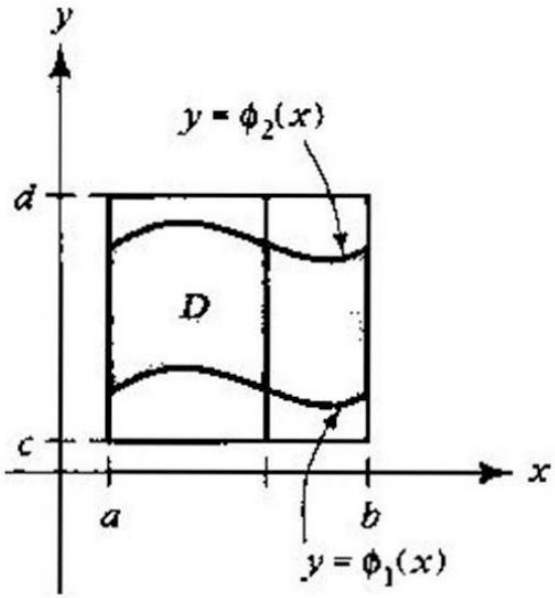
$$\iint_D f(x, y)\, \mathrm{d}x\mathrm{d}y = \iint_R \bar{f}(x, y)\, \mathrm{d}x\mathrm{d}y = \int_a^b \int_c^d \bar{f}(x, y)\, \mathrm{d}y\mathrm{d}x$$

$$= \int_c^d \int_a^b \bar{f}(x, y)\, \mathrm{d}x\mathrm{d}y$$

where $\bar{f}$ equals f in D and is zero outside D.

If D is a region of Type 1 determined by functions ϕ_1 and ϕ_2 on $[a, b]$, the integral $\int_c^d \bar{f}(x, y)\ \mathrm{d}y$ becomes (see Figure 7.6):

$$\int_c^d \bar{f}(x, y)\ \mathrm{d}y = \int_{\phi_1(x)}^{\phi_2(x)} \bar{f}(x, y)\ \mathrm{d}y = \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\ \mathrm{d}y \tag{7.2}$$



Equation (7.2) hold since $\bar{f}(x, y) = f(x, y)$ for $\phi_1(x) \le y \le \phi_2(x)$ and $\bar{f}(x, y) = 0$ otherwise. So

$$\iint_D f(x, y)\ \mathrm{d}x\mathrm{d}y = \int_a^b \left[\int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\ \mathrm{d}y \right] \mathrm{d}x$$

Iterated Integrals for Elementary Regions

Using a similar calculation for Type 2 regions, we get the second formula in the following box.

Theorem 7.3.7

If D is a region of Type 1

$$\iint_D f(x,y)\,\mathrm{d}x\mathrm{d}y = \int_a^b \left[\int_{\phi_1(x)}^{\phi_2(x)} f(x,y)\,\mathrm{d}y\right]\mathrm{d}x$$

If D is of Type 2

$$\iint_D f(x,y)\,\mathrm{d}x\mathrm{d}y = \int_c^d \left[\int_{\psi_1(y)}^{\psi_2(y)} f(x,y)\,\mathrm{d}x\right]\mathrm{d}y$$

If D is of both types, either formula is applicable.

Interpret iterated integration as Cavalieri's principle

Suppose $f(x, y) \geq 0$. Iterated integration implements **Cavalieri's principle** for computing volumes by slicing.

- **For Type 1 Regions:** (D defined by $\phi_1(x) \leq y \leq \phi_2(x)$ on $[a, b]$)
  1. Fix x and slice with plane through $(x, 0, 0)$ parallel to yz-plane
  2. Obtain the region in the yz-plane defined by the inequality $0 \leq z \leq f(x, y)$
  3. The cross-section has area:

$$A(x) = \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\ \mathrm{d}y$$

  4. Total volume equals:

$$\iint_D f(x, y)\ \mathrm{d}x\mathrm{d}y = \int_a^b A(x)\ \mathrm{d}x = \int_a^b \left(\int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\ \mathrm{d}y \right) \mathrm{d}x$$

Continue

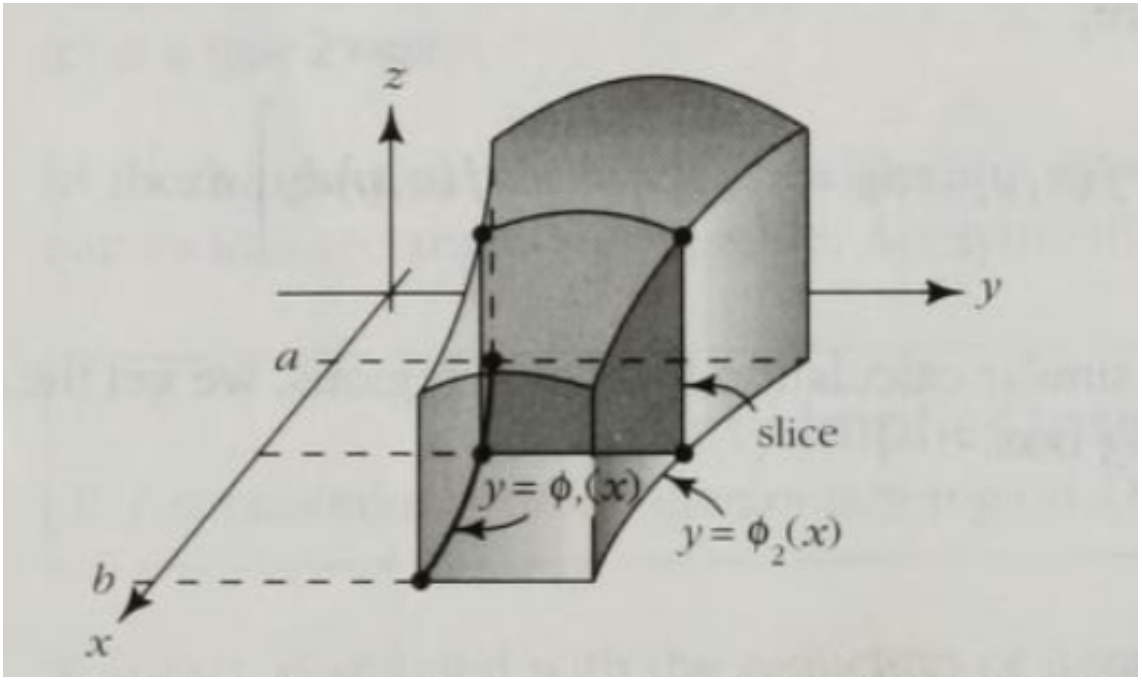


Figure 7.6: The area of the slice with fixed x is $\int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\ \mathrm{d}y$.

- **For Type 2 Regions**:
  - Analogous procedure uses slices parallel to xz-plane
  - Yields the alternative iterated integral form:

$$\iint_D f(x,y)\ \mathrm{d}x\mathrm{d}y = \int_c^d \left(\int_{\psi_1(y)}^{\psi_2(y)} f(x,y)\ \mathrm{d}x \right) \mathrm{d}y$$

Example 7.3.8

Let D be the region bounded by the graphs of

$$y = \sin x \text{ and } y = 0 \quad \text{for} \quad 0 \leq x \leq 2\pi$$

Find the iterated integral of the constant function $f(x, y) = 7$ over D.

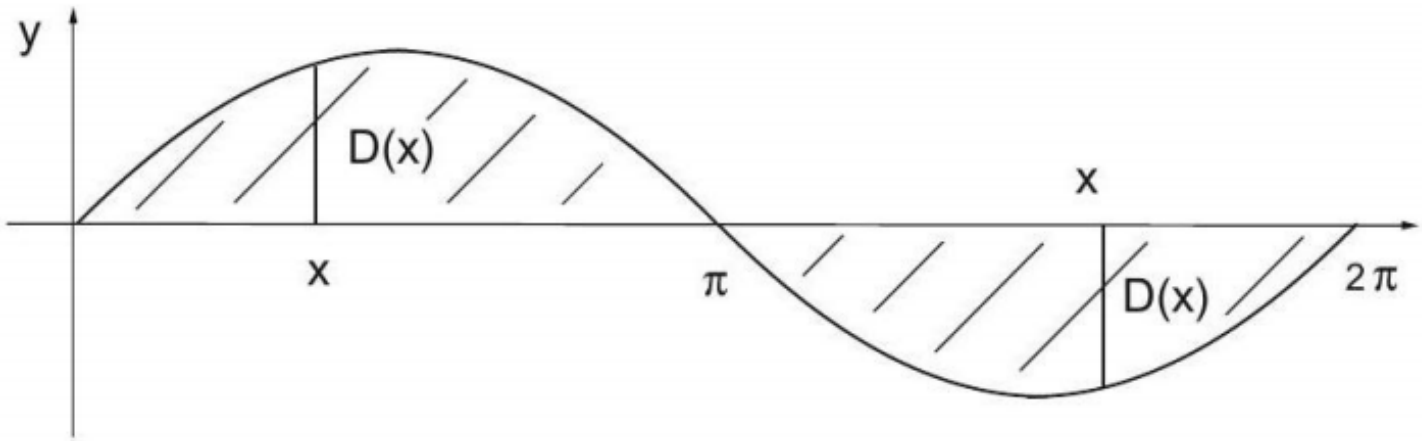


Figure 7.7: The set D in Example 7.3.8

Solution

We sketch the region D in Figure 7.8.

- D is the union of two type 1 regions.
- For $0 \leq x \leq \pi$, we have $0 \leq y \leq \sin x$, and for $\pi \leq x \leq 2\pi$ we have $\sin x \leq y \leq 0$.

So we set up two iterated integrals:

$$\int_{x=0}^{x=\pi} \left(\int_{y=0}^{y=\sin x} 7 \, \mathrm{d}y \right) \mathrm{d}x + \int_{x=\pi}^{x=2\pi} \left(\int_{y=\sin x}^{y=0} 7 \, \mathrm{d}y \right) \mathrm{d}x$$

$$= \int_{x=0}^{x=\pi} 7 \sin x \, \mathrm{d}x + \int_{x=\pi}^{x=2\pi} -7 \sin x \, \mathrm{d}x$$

$$= [-7 \cos x]_{x=0}^{x=\pi} + [7 \cos x]_{x=\pi}^{x=2\pi}$$

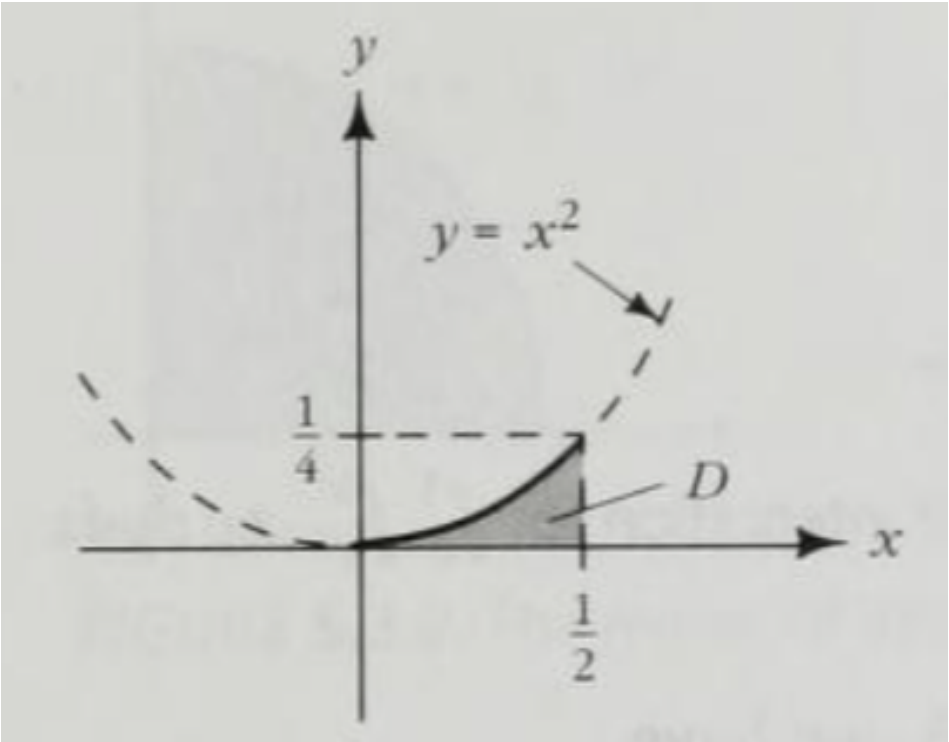
$$= -7(-1-1) + 7(1-(-1)) = 28.$$

Example 7.3.9

Find

$$\iint_D (x+y)\,\mathrm{d}x\mathrm{d}y$$

where D is the shaded region in the following figure.



Solution

- D is a region of Type 1, with $[a,b] = \left[0, \frac{1}{2}\right], \phi_1(x) = 0$, and $\phi_2(x) = x^2$.

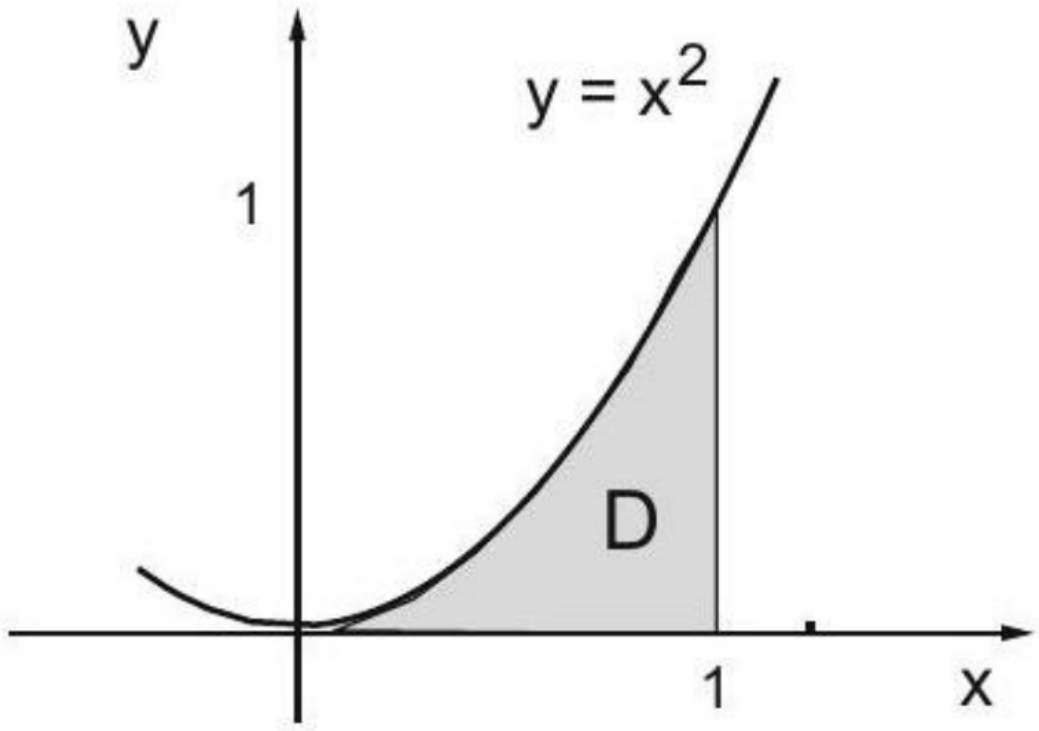
  By Theorem 7.3.7,

$$\begin{aligned}
\iint_D (x+y)\,\mathrm{d}x\mathrm{d}y &= \int_0^{1/2} \int_0^{x^2} (x+y)\,\mathrm{d}y\mathrm{d}x \\
&= \int_0^{1/2} \left[\left(xy + \frac{y^2}{2} \right) \Big|_{y=0}^{x^2} \right] \mathrm{d}x \\
&= \int_0^{1/2} \left(x^3 + \frac{x^4}{2} \right) dx = \left(\frac{x^4}{4} + \frac{x^5}{10} \right) \Big|_0^{1/2} \\
&= \frac{1}{64} + \frac{1}{320} = \frac{3}{160}
\end{aligned}$$

- D is also a region of Type 2 , with $[c,d] = \left[0, \frac{1}{4}\right], \psi_1(y) = \sqrt{y}$, $\psi_2(y) = \frac{1}{2}$.
  - We leave it to you to verify that the double integral calculated by the second formula in Theorem 7.3.7 is also $3/160$.

Example 7.3.10

Compute the double integral of $f(x, y) = xy^3$ over the domain D shown in the figure below.



Solution

- D is bounded by the lines $y = 0, y = 1$, and the graphs of the functions (x in terms of y) $x = 1$ and $x = \sqrt{y}$.

  Then

$$\int_D xy^3 \, \mathrm{d}A = \int_0^1 \left(\int_{\sqrt{y}}^1 xy^3 \, \mathrm{d}x \right) \mathrm{d}y$$

$$= \int_0^1 \frac{1}{2} \left(y^3 - y^4 \right) \mathrm{d}y = \frac{1}{2} \left(\frac{1}{4} - \frac{1}{5} \right) = \frac{1}{40}$$

- This double integral may also be evaluated by

$$\int_D xy^3 \, \mathrm{d}A = \int_0^1 \left(\int_{y=0}^{y=x^2} xy^3 \, \mathrm{d}y \right) \mathrm{d}x$$

$$= \int_0^1 x \left(\frac{1}{4} \left(x^2 \right)^4 \right) - \frac{1}{4}(0)^2 \mathrm{d}x = \int_0^1 \frac{1}{4} x^9 \, \mathrm{d}x = \frac{1}{40}$$

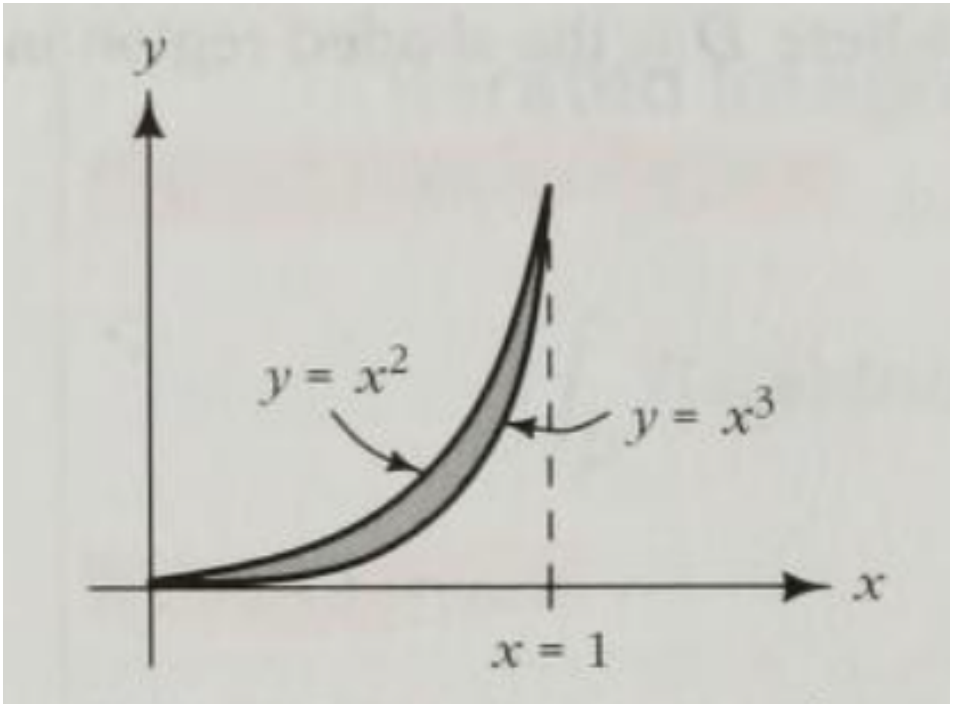
Example 7.3.11

Evaluate

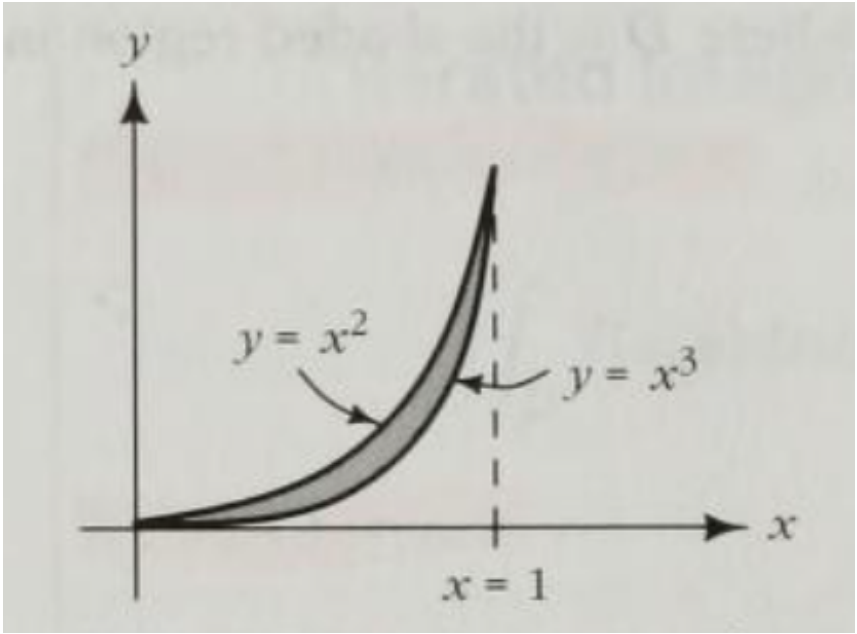
$$\int_0^1 \int_{x^3}^{x^2} xy \, \mathrm{d}y\mathrm{d}x$$

Sketch the region D for which this iterated integral gives the double integral

$$\iint_D xy \, \mathrm{d}x\mathrm{d}y$$



Solution



Here y ranges from x^3 to x^2, while x goes from 0 to 1.

- The integral is

$$\int_0^1 \left(\frac{xy^2}{2}\Big|_{y=x^3}^{x^2} \right) \mathrm{d}x = \int_0^1 \left(\frac{x^5}{2} - \frac{x^7}{2} \right) \mathrm{d}x$$

$$= \left(\frac{x^6}{12} - \frac{x^8}{16} \right)\Big|_{x=0}^{1} = \frac{1}{12} - \frac{1}{16} = \frac{1}{48}$$

Note: If a region D is of Type 3 , we have

$$\iint_D f(x,y)\,\mathrm{d}x\mathrm{d}y = \int_a^b \int_{\phi_2(x)}^{\phi_2(x)} f(x,y)\,\mathrm{d}y\mathrm{d}x = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x,y)\,\mathrm{d}x\mathrm{d}y$$

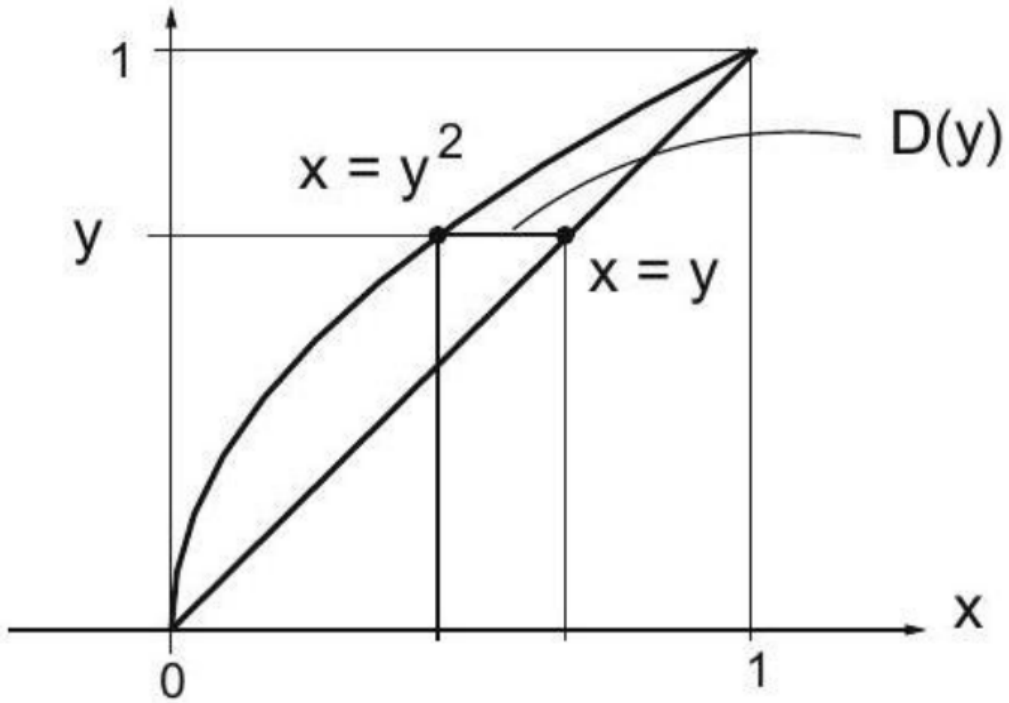
- The equality of these expressions allows **changing integration order**
- Useful when one order proves difficult to integrate
- Provides flexibility in computation approaches

Example 7.3.12

Let D be the region bounded by the graphs of

$$x = y^2 \quad \text{and} \quad x = y$$

Find the iterated integral of $f(x, y) = 2xy^3$.



Solution

The region D is of type 3. Viewing D as a type 2 region:

- Every point (x, y) in D has $0 \leq y \leq 1$, and
- for each y between 0 and 1, $\psi_1(y) = y^2$ and $\psi_2(y) = y$

The iterated integral:

$$\int_0^1 \left(\int_{\psi_1(y)}^{\psi_2(y)} 2xy^3 \, \mathrm{d}x \right) \mathrm{d}y = \int_0^1 \left(\int_{y^2}^{y} 2xy^3 \, \mathrm{d}x \right) \mathrm{d}y = \int_0^1 \left[x^2 y^3 \right]_{x=y^2}^{x=y} \, \mathrm{d}y$$

$$= \int_0^1 \left(y^5 - y^7 \right) \mathrm{d}y = \frac{1}{6} - \frac{1}{8} = \frac{1}{24}$$

Solution (Continued)

Viewing D as a type 1 region, we see that

- every point in D has $0 \leq x \leq 1$.
- for each x between 0 and 1, $\phi_1(x) = x$ and $\phi_1(x) = \sqrt{x}$.

The iterated integral:

$$\int_0^1 \left(\int_{\phi_1(x)}^{\phi_2(x)} 2xy^3 \, \mathrm{d}y \right) \mathrm{d}x = \int_0^1 \left(\int_x^{\sqrt{x}} 2xy^3 \, \mathrm{d}y \right) \mathrm{d}x = \int_0^1 \left[\frac{1}{2} x y^4 \right]_{y=x}^{y=\sqrt{x}} \mathrm{d}x$$

$$= \int_0^1 \left(\frac{1}{2}x^3 - \frac{1}{5}x^5 \right) \mathrm{d}x = \frac{1}{8} - \frac{1}{12} = \frac{1}{24}$$

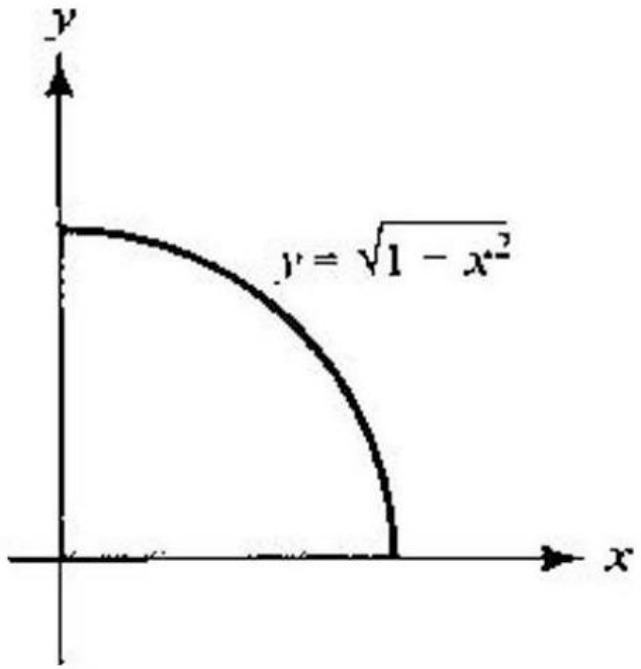
Example 7.3.13

Write

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \sqrt{1-y^2}\, \mathrm{d}y\mathrm{d}x$$

as an integral over a region.

Sketch the region and show that it is of Types 1 and 2. Reverse the order of integration and evaluate.



Solution

Region D is of:

- Type 1 with $\phi_1(x) = 0,\ \phi_2(x) = \sqrt{1-x^2}$
- Type 2 with $\psi_1(y) = 0,\ \psi_2(y) = \sqrt{1-y^2}$

Thus

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \sqrt{1-y^2}\, \mathrm{d}y\mathrm{d}x = \int_0^1 \int_0^{\sqrt{1-y^2}} \sqrt{1-y^2}\, \mathrm{d}x\mathrm{d}y$$

$$= \int_0^1 \left[\sqrt{1-y^2}x \Big|_{x=0}^{\sqrt{1-y^2}} \right] \mathrm{d}y$$

$$= \int_0^1 \left(1-y^2\right)\ \mathrm{d}y = \left(y - \frac{y^3}{3} \right) \Big|_0^1 = \frac{2}{3}$$

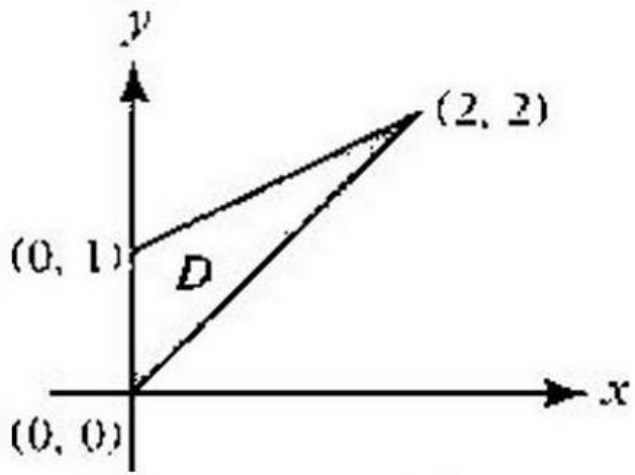
Observation: Evaluating the integral in the original order is much harder!

Example 7.3.14

Calculate the integral of

$$f(x, y) = (x + y)^2$$

over the region shown in the figure below.



Remark:

In this example, the region is of Type 1 but not of Type 2. There is a preferred order of integration.

We could use the other order of integration, i.e., x first, but this requires us to break up the region into two regions of Type 2 by drawing the line $y = 1$

Solution

Optimal integration order can perform the integration in one step:

$$\iint_D f(x,y)\,\mathrm{d}x\mathrm{d}y = \int_0^2 \int_x^{\frac{1}{2}x+1} f(x,y)\,\mathrm{d}y\mathrm{d}x$$

(The lines bounding D on the bottom and top are $y = x$ and $y = \frac{1}{2}x + 1$)

- The integral is thus

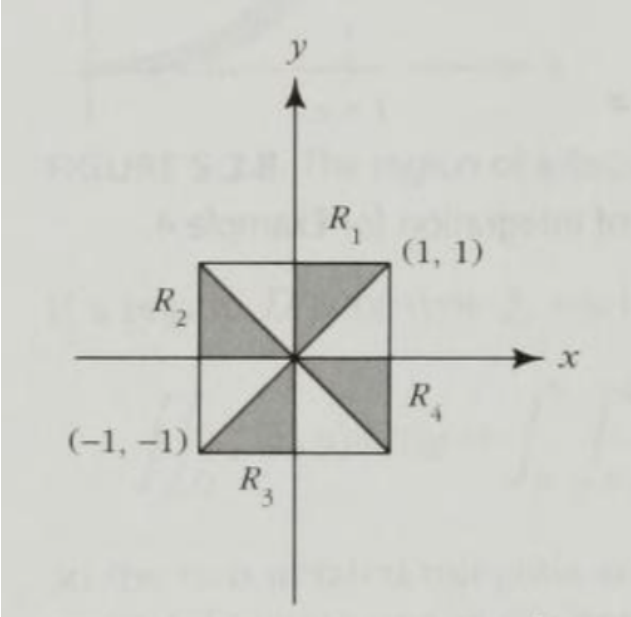
$$\begin{aligned}
\int_0^2 \int_x^{\frac{1}{2}x+1} (x+y)^2\,\mathrm{d}y\mathrm{d}x &= \int_0^2 \left[\frac{1}{3}(x+y)^3\Big|_{y=x}^{\frac{1}{2}x+1}\right]\mathrm{d}x \\
&= \frac{1}{3}\int_0^2 \left[\left(\frac{3}{2}x+1\right)^3 - (2x)^3\right]\mathrm{d}x \\
&= \frac{1}{3}\left[\frac{1}{6}\left(\frac{3}{2}x+1\right)^4\Big|_0^2 - 2x^4\Big|_0^2\right] \\
&= \frac{1}{3}\left[\frac{1}{6}\left(4^4-1\right) - 2\cdot 16\right] = \frac{1}{3}\left[\frac{21}{2}\right] = \frac{7}{2}
\end{aligned}$$

Example 7.3.15

Find

$$\iint_D x^2 \, \mathrm{d}x\mathrm{d}y$$

where D is the shaded region in the figure below.



Remark:

General regions can often be broken into elementary regions, and double integrals over these regions can be computed one piece at a time.

Solution

Each of the regions R_1, R_2, R_3, R_4 is of types 1 and 2, so we may integrate over each one separately and sum the results:

$$\iint_{R_1} x^2 \, \mathrm{d}x\mathrm{d}y = \int_0^1 \int_x^1 x^2 \, \mathrm{d}y\mathrm{d}x = \int_0^1 \left(x^2 y\right)\Big|_{y=x}^1 \mathrm{d}x$$

$$= \int_0^1 \left(x^2 - x^3\right) \, \mathrm{d}x = \left(\frac{x^3}{3} - \frac{x^4}{4}\right)\Big|_0^1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

Similarly, we find

$$\iint_{R_2} x^2 \, \mathrm{d}x\mathrm{d}y = \int_{-1}^0 \int_0^{-x} x^2 \, \mathrm{d}y\mathrm{d}x = \frac{1}{4}$$

Solution (Continued)

Since the integrand is unchanged under reflection through the origin, the integrals over R_1 and R_3 are equal, as are those over R_2 and R_4. Therefore,

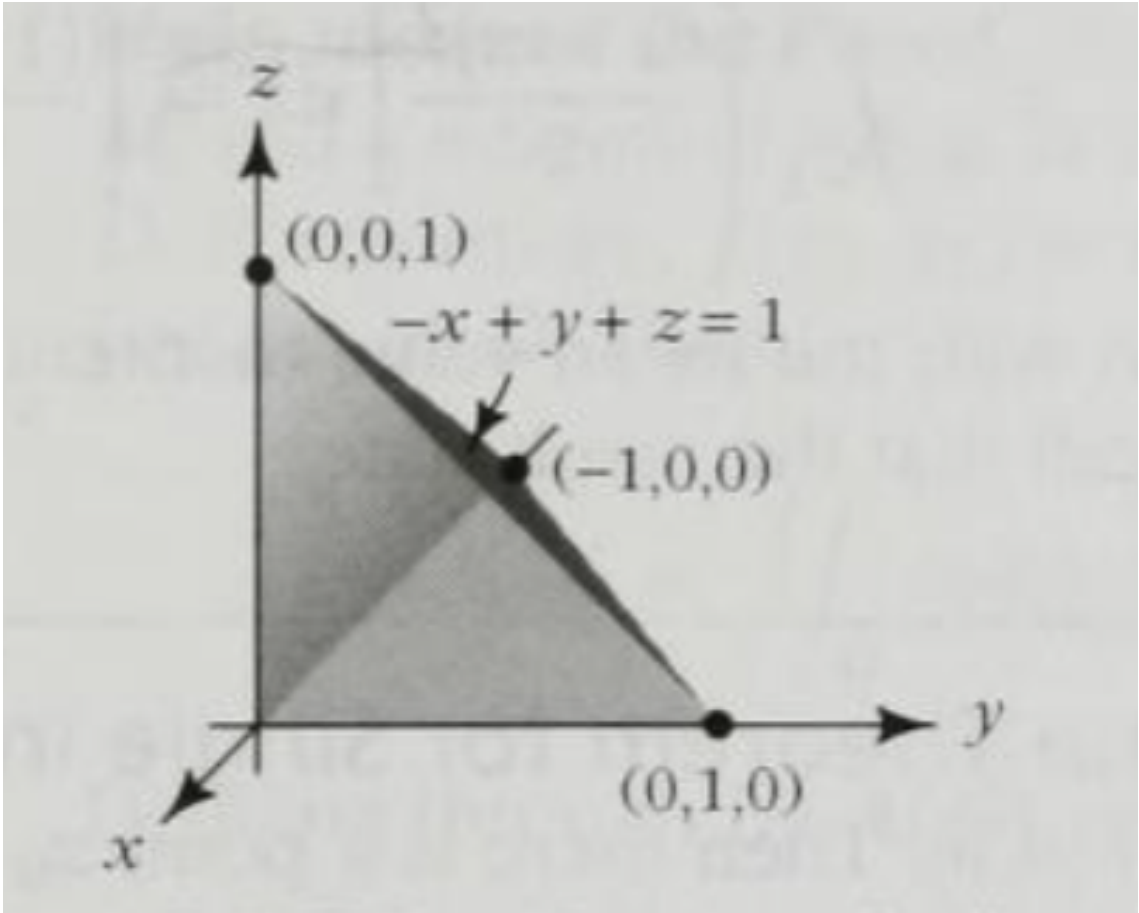
$$\iint_{R_3} x^2 \, \mathrm{d}x\mathrm{d}y = \frac{1}{12} \quad \text{and} \quad \iint_{R_4} x^2 \, \mathrm{d}x\mathrm{d}y = \frac{1}{4}$$

so adding these:

$$\iint_{D} x^2 \, \mathrm{d}x\mathrm{d}y = \frac{1}{12} + \frac{1}{4} + \frac{1}{12} + \frac{1}{4} = \frac{2}{3}.$$

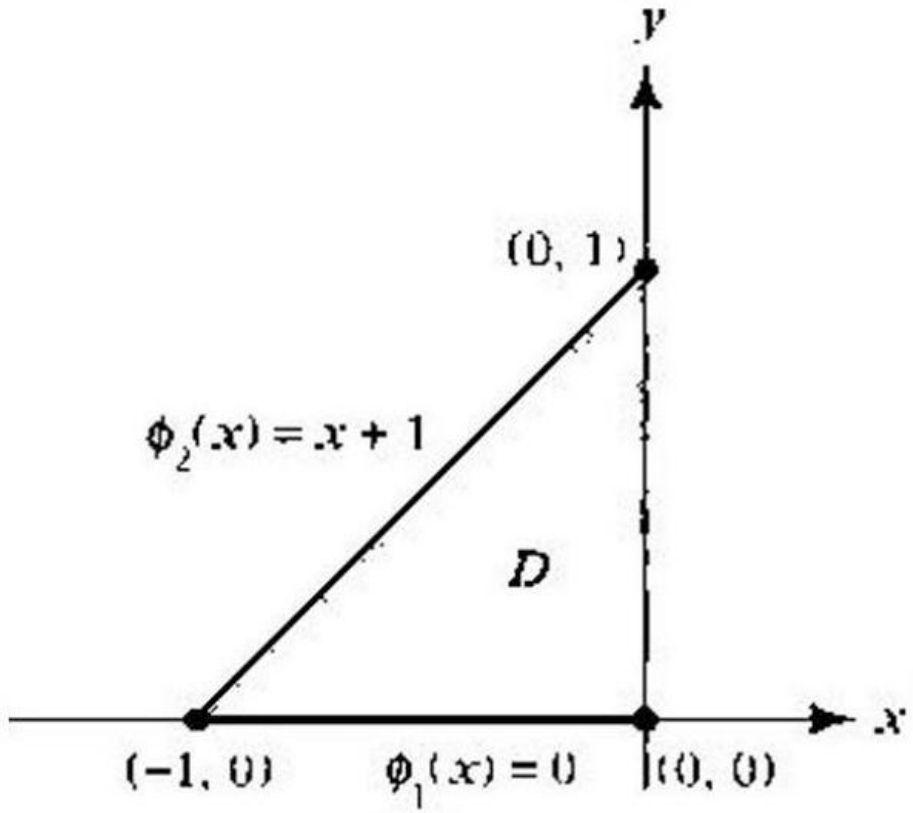
Example 7.3.16

Find the volume of the tetrahedron bounded by the planes $y = 0, z = 0, x = 0$, and $-x + y + z = 1$.



Solution

The tetrahedron has a triangular base D, whose points (x, y) satisfy $-1 \leq x \leq 0$ and $0 \leq y \leq 1 + x$; hence D is a region of Type 1.



Solution (Continued)

For any point (x, y) in D, the height of the surface z above (x, y) is $1 - y + x$.

Thus, the volume we seek is the double integral

$$\iint_D (1 - y + x)\, \mathrm{d}x\mathrm{d}y$$

Integrating with respect to y first, we get

$$\begin{aligned}
\iint_D (1 - y + x)\, \mathrm{d}x\mathrm{d}y &= \int_{-1}^{0} \int_{0}^{1+x} (1 - y + x)\, \mathrm{d}y\mathrm{d}x \\
&= \int_{-1}^{0} \left[(1 + x)y - \frac{y^2}{2} \right]_{y=0}^{1+x} \mathrm{d}x \\
&= \int_{-1}^{0} \left[\frac{(1 + x)^2}{2} \right] \mathrm{d}x = \left[\frac{(1 + x)^3}{6} \right]_{-1}^{0} = \frac{1}{6}
\end{aligned}$$

Section 4

Change of variables in a double integral

A change of variables relates two pairs of variables (x, y) and (u, v) by a function $\mathbf{F}$ from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Definition 7.4.1

A continuously differentiable function

$$\mathbf{F}(u, v) = (x(u, v), y(u, v))$$

from an open set U in the plane to $\mathbb{R}^2$ is called a **smooth change of variables** if

- $\mathbf{F}$ is one to one and
- its derivative matrix is invertible at each point of U.

Jacobian Determinant

If $\mathbf{F}$ is a smooth change of variables then the derivative matrix and its determinant:

$$J\mathbf{F}(u,v) = \det D\mathbf{F}(u,v) = \det \begin{bmatrix} x_u & x_v \\ y_u & y_v \end{bmatrix}$$

- is always nonzero for smooth changes
- its absolute value represents local area magnification factor at (u,v) of area under the mapping

1. Consider triangle S with vertices:

$$(u,v), \quad (u+h,v), \quad (u,v+h)$$

These points are mapped into the points

$$(x(u,v),y(u,v)), \quad (x(u+h,v),y(u+h,v)), \quad (x(u,v+h),y(u,v+h))$$

in the x,y plane and are vertices of a triangle T.

Continue

2. We approximate:

$$(x(u+h,v), y(u+h,v)) \approx (x + hx_u, y + hy_u)$$
$$(x(u,v+h), y(u,v+h)) \approx (x + hx_v, y + hy_v)$$

where the functions x, y and their partial derivatives are evaluated at (u, v).

For small h, linear approximation gives:

$T' \approx$ triangle with vertices $(x, y),\ (x + hx_u, y + hy_u),\ (x + hx_v, y + hy_v)$

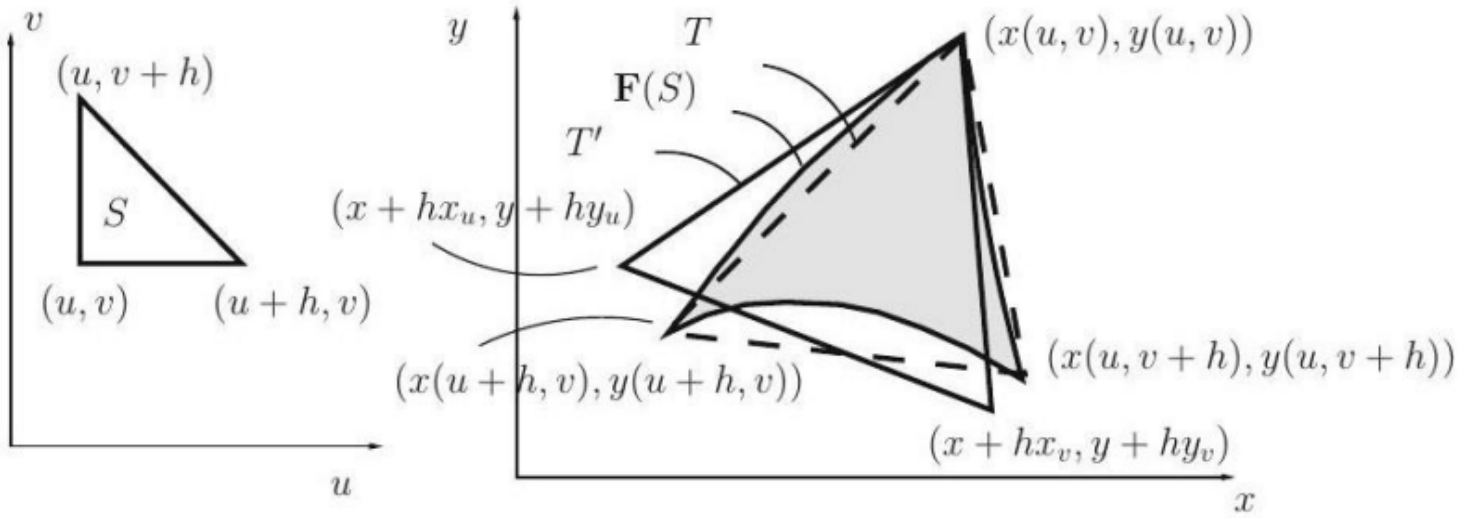


Figure 7.8: $\text{Area}(T) - \text{Area}\,(T')$ is less than a constant multiple of h^3.

Continue

3. By a careful error analysis, we can establish

$$0 \leq \frac{\text{Area}(T) - \text{Area}(T')}{\text{Area}(S)} \leq \frac{kh^3}{h^2/2} \xrightarrow{h \to 0} 0$$

Thus:

$$\lim_{h \to 0} \frac{\text{Area}(T')}{\text{Area}(S)} = \lim_{h \to 0} \frac{\text{Area}(T)}{\text{Area}(S)}$$

Continue

4. Exact Area Calculation:

$$\text{Area}(T') = \left|\frac{1}{2}(hx_u hy_v - hx_v hy_u)\right| = \frac{1}{2}|x_u y_v - x_v y_u|h^2$$

$$\text{Area}(S) = \frac{1}{2}h^2$$

$$\frac{\text{Area}(T')}{\text{Area}(S)} = |J\mathbf{F}(u,v)| = |x_u y_v - x_v y_u| \tag{7.3}$$

Here is the change of variables theorem for integrals.

Theorem 7.4.2

Let $F(u,v) = (x(u,v), y(u,v))$ denote a smooth change of variables mapping:

- *A bounded set C with smooth boundary onto D*
- *∂C (boundary of C) is mapped to ∂D(boundary of D)*

For continuous f on D:

$$\int_D f(x,y)\ \mathrm{d}x\mathrm{d}y = \int_C f(x(u,v), y(u,v))|J\mathbf{F}(u,v)|\ \mathrm{d}u\mathrm{d}v \tag{7.4}$$

where $J\mathbf{F}$ is the Jacobian of the mapping.

Sketch of proof

1. Riemann Sum Approximation:

   Divide the u, v plane into h-squares and approximate

   $$\int_C f(x(u,v), y(u,v)) |J\mathbf{F}(u,v)| \, \mathrm{d}u\mathrm{d}v$$

   by a finite sum

   $$S(f, C, h) = \sum_j f\left(\mathbf{F}\left(u_j, v_j\right)\right) \left|J\mathbf{F}\left(u_j, v_j\right)\right| h^2$$

   where (u_j, v_j) is some point in the j-th h-square, and the sum taken over all h-squares contained in C.

2. The mapping $\mathbf{F}$ carries:
   - h-squares in the u, v plane $\to$ smoothly bounded sets in the x, y plane, D_j
   - Area$(D_j) \approx |J\mathbf{F}(u_j, v_j)|h^2$ with an error less than some multiple of h^3

Proof (Continued)

3. Let D' be the union of the sets D_j. The integral of f over D' is the sum of the integrals of f over D_j :

$$\int_{D'} f \, \mathrm{d}x\mathrm{d}y = \sum_j \int_{D_j} f \, \mathrm{d}x\mathrm{d}y$$

4. Approximation Errors: (each term)

$$\int_{D_j} f \, \mathrm{d}x\mathrm{d}y \approx f(u_j, v_j)\mathsf{Area}(D_j)$$

$$\mathsf{Area}(D_j) \approx |J\mathbf{F}\,(u_j, v_j)|\, h^2$$

The error thus committed in each term in the sum is $< ch^3$, for some constant c.

The total error committed is $\leq$ the sum of the individual errors.

Proof (Continued)

5. Since the number of terms in the sum is $< \frac{\text{Area}(D)}{h^2}$, the total error satisfies:

$$\text{Total error} < \frac{\text{Area}(D)}{h^2} \cdot ch^3 = \text{Area}(D)ch$$

Thus:

$$\left| S(f, C, h) - \int_{D'} f \, \mathrm{d}A \right| \leq Ch$$

So $\int_{D'} f(x, y) \, \mathrm{d}A$ and $\int_C f(x(u, v), y(u, v))|J\mathbf{F}(u, v)| \, \mathrm{d}u\mathrm{d}v$ differ $<$ some constant times h.

6. Limit Behavior: As $h \to 0$
   - The area of the set of points that are in D but not in D' does not exceed Mh, where $M > 0$ is a constant depending on the boundary of C and the upper bound of $|J\mathbf{F}(u, v)|$ on the closure of C.
   - the difference between $\int_{D'} f(x, y) \, \mathrm{d}A$ and $\int_D f(x, y) \, \mathrm{d}A$ tends to zero as h tends to zero

Example 7.4.3

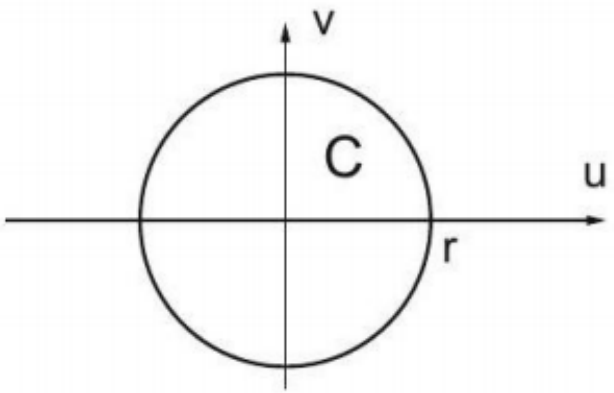
Define sets

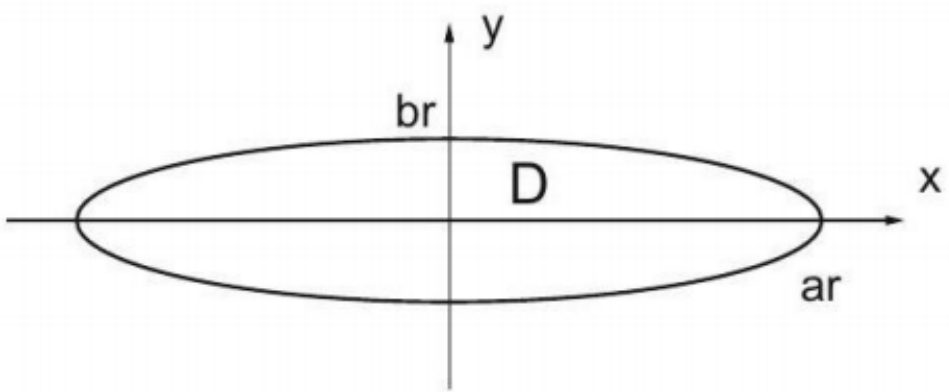
$$C = \left\{(u, v) : u^2 + v^2 \leq r^2\right\}, \quad D = \left\{(x, y) : \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 \leq r^2\right\}$$

The boundary of C is a circle and the boundary of D is an ellipse.

The mapping $\mathbf{F}(u, v) = (au, bv)$, $x = au$, $y = bv$, $a > 0$, $b > 0$ sends points (u, v) in the set C one to one onto the points of D.

Find the area of D.





Solution

1. Compute Jacobian:

$$J\mathbf{F}(u,v) = \det D\mathbf{F} = \det \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = ab > 0$$

2. Apply change of variables:

$$\text{Area}(D) = \int_D 1 \ \mathrm{d}x\mathrm{d}y = \int_C ab \ \mathrm{d}u\mathrm{d}v$$

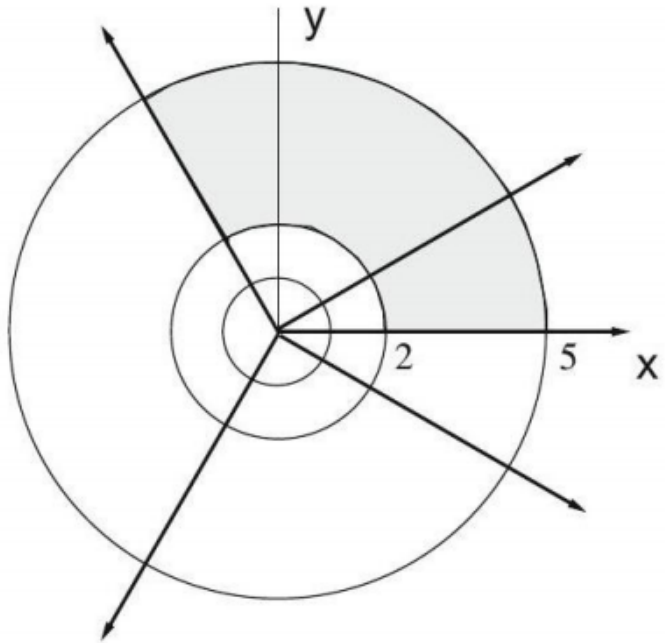
3. Calculate circular integral:

$$ab \int_{u^2+v^2 \le r^2} \mathrm{d}u\mathrm{d}v = ab\pi r^2$$

Example 7.4.4

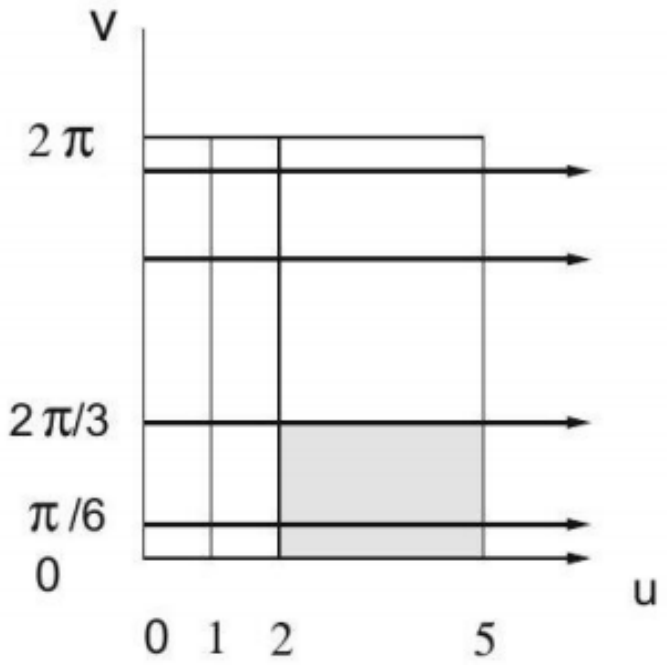
Let D be the shaded set in the figure below. Find

$$\int_D e^{x^2+y^2}\, dA$$



Hint: The iterated integrals we would obtain in x, y coordinates will not help us evaluate this integral because we cannot find a simple anti-derivative for $e^{x^2+y^2}$ with respect to x or y.

Solution



We try a change of variables $\mathbf{F}$ that is suggested by the polar coordinate transformation

$$x = u\cos v, \quad y = u\sin v.$$

- Note that $\mathbf{F}$ is one to one on an open set that contains the rectangle C given by $2 \le u \le 5$, $0 \le v \le 2\pi/3$.

Solution cont'd

1. The Jacobian is

$$J\mathbf{F}(u,v) = \det D\mathbf{F}(u,v) = \det \begin{bmatrix} \cos v & -u\sin v \\ \sin v & u\cos v \end{bmatrix} = u\cos^2 v + u\sin^2 v = u$$

   - The Jacobian is positive in C.

2. By the change of variables theorem:

$$\int_D \mathrm{e}^{x^2+y^2}\,\mathrm{d}x\mathrm{d}y = \int_C \mathrm{e}^{u^2\cos^2 v + u^2\sin^2 v} u\,\mathrm{d}u\mathrm{d}v = \int_{v=0}^{v=2\pi/3} \left(\int_{u=2}^{u=5} \mathrm{e}^{u^2} u\,\mathrm{d}u \right) \mathrm{d}v$$

$$= \int_{v=0}^{v=2\pi/3} \frac{1}{2}\left(\mathrm{e}^{25} - \mathrm{e}^4\right) \mathrm{d}v = \frac{1}{2}\left(\mathrm{e}^{25} - \mathrm{e}^4\right)\frac{2\pi}{3}$$

Remark

It is customary to write (r, θ) for polar coordinates instead of (u, v). Then:

The change of variables formula for using **polar coordinates** is

$$\int_D f(x, y)\,\mathrm{d}x\mathrm{d}y = \int_C f(r\cos\theta, r\sin\theta) r\,\mathrm{d}r\mathrm{d}\theta$$

Example 7.4.5

Evaluate

$$\int_D e^{x^2+y^2}\,\mathrm{d}A$$

where $D = \{(x, y) : x^2 + y^2 \leq 25\}$ and let C denote the rectangle $0 \leq u \leq 5$, $0 \leq v \leq 2\pi$.

Solution

The polar coordinate mapping $\mathbf{F}(u,v) = (u\cos v, u\sin v)$ is not a smooth change of variables on a set that contains $C : \det D\mathbf{F}(0,0) = 0$ and also $\mathbf{F}$ is not one to one, $\mathbf{F}(u,0) = \mathbf{F}(u,2\pi)$.

- For $0 < \epsilon$ let
$$C_\epsilon = \{(u,v) : \epsilon \le u \le 5, 0 \le v \le 2\pi - \epsilon\}$$
and set $D_\epsilon = \mathbf{F}(C_\epsilon)$.
- The integrand $\mathrm{e}^{x^2+y^2}$ is bounded on D and the area of the region $C \setminus C_\epsilon \to 0$ as $\epsilon \to 0$, as does the area of the region $D \setminus D_\epsilon$. Then
$$\int_{D_\epsilon} \mathrm{e}^{x^2+y^2}\,\mathrm{d}x\mathrm{d}y = \int_{C_\epsilon} \mathrm{e}^{u^2} u\,\mathrm{d}u\mathrm{d}v \;\Rightarrow \int_D \mathrm{e}^{x^2+y^2}\,\mathrm{d}x\mathrm{d}y = \int_C \mathrm{e}^{u^2} u\,\mathrm{d}u\mathrm{d}v$$
- We evaluate the last integral as an iterated integral
$$= \int_{v=0}^{v=2\pi} \left(\int_{u=0}^{u=5} \mathrm{e}^{u^2} u\,\mathrm{d}u \right) \mathrm{d}v = \int_{v=0}^{v=2\pi} \left[\frac{1}{2}\mathrm{e}^{u^2} \right]_{u=0}^{u=5} \mathrm{d}v = \frac{1}{2}\left(\mathrm{e}^{25} - 1\right) 2\pi$$

Example 7.4.6

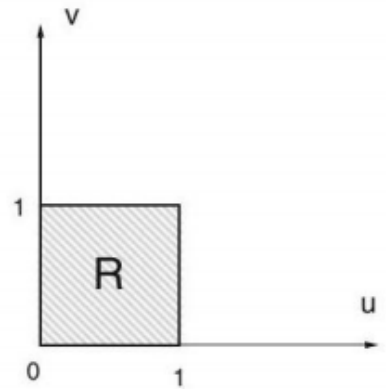
Let D be the parallelogram in $\mathbb{R}^2$ bounded by the lines

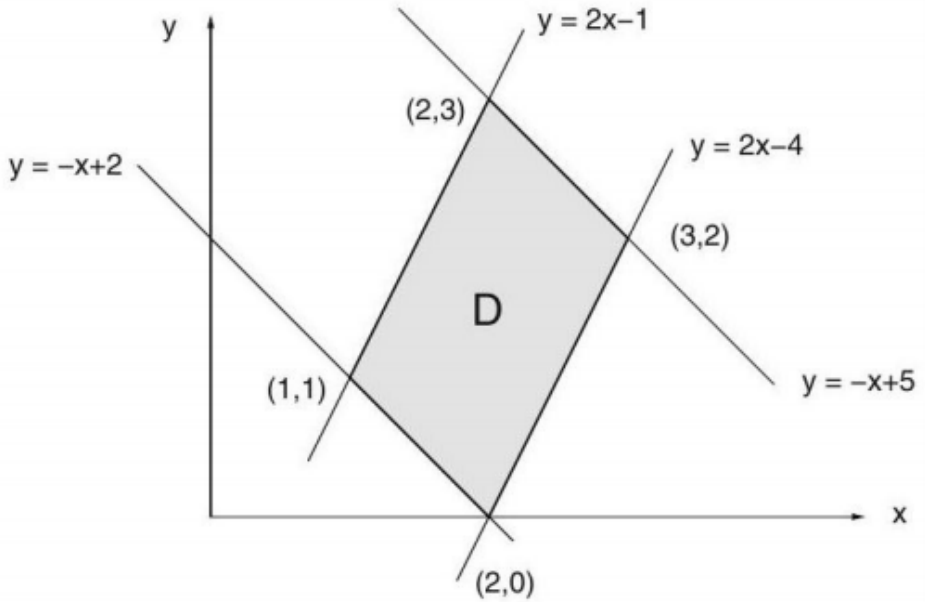
$$y = -x + 5, \quad y = -x + 2, \quad y = 2x - 1, \quad y = 2x - 4$$

and let R be the unit square $[0, 1] \times [0, 1]$.

Compute

$$\int_D e^x \, dA$$





Solution

1. The mapping $\mathbf{F}(u,v) = (u+v+1, -u+2v+1)$ takes R one to one onto the parallelogram D.

2. Compute Jacobian:

$$D\mathbf{F}(u,v) = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}, \quad J\mathbf{F}(u,v) = \det(D\mathbf{F}(u,v)) = 3$$

3. Transform and evaluate:

$$\begin{aligned} \int_D \mathrm{e}^x \, \mathrm{d}x\mathrm{d}y &= \int_R \mathrm{e}^{u+v+1}|3| \, \mathrm{d}u\mathrm{d}v = 3\mathrm{e}\int_0^1\int_0^1 \mathrm{e}^u\mathrm{e}^v \, \mathrm{d}u\mathrm{d}v \\ &= 3\mathrm{e}\int_0^1 \mathrm{e}^u \, \mathrm{d}u \int_0^1 \mathrm{e}^v \, \mathrm{d}v = 3\mathrm{e}(\mathrm{e}-1)^2 \end{aligned}$$